

Chapter 3

Method (additions)

3.6 Evaluation Protocol for Automatic Prompting

All automatic-prompting experiments (Section 4.3) share one protocol, so that every row of every table is directly comparable. The protocol fixes the frozen model, the data split, the training recipe of the learned prompters, the conversion of a prompter output into SAM prompts and the evaluation metrics.

Frozen model. Phase 2 uses one frozen Phase-1 model, the Infused-Token ViT-B checkpoint with MLP ratio 1.0 (Table 4.12: TC 0.724 / AR 0.632 IoU with ground-truth boxes). The Infused-Token checkpoint with MLP ratio 0.5 serves as a robustness check. Input adapter, encoder and decoder are not changed, except in the decoder-adaptation experiments (Section 3.8).

Data split. The 398 ClimateNet training images are split once into 358 training and 40 validation images (fixed random permutation). The validation images are used only to select the checkpoint of each learned prompter and to tune post-processing parameters; the 61 test images are used once per final model. (The original Phase-2 scripts selected the best epoch on the test set, which biases their numbers upwards.)

Feature cache. Because the encoder is frozen and no augmentation is used (Section 2.2.2), the encoder is run once per image and its outputs are cached: the neck embedding $E \in \mathbb{R}^{256 \times 64 \times 64}$, the outputs of all twelve ViT blocks $F_1, \dots, F_{12} \in \mathbb{R}^{768 \times 64 \times 64}$ and the four CG-Net input fields. This is numerically identical to running the encoder in every training step and reduces a 60-epoch training of a small prompter from hours to minutes.

Training of learned prompters. Every learned prompter outputs two logit maps $\ell_{\text{TC}}, \ell_{\text{AR}}$ (independent sigmoids) and is trained with the loss selected in Section 4.1 (Table 4.4), summed over both classes,

$$\mathcal{L}_{\text{seg}} = \sum_{c \in \{\text{TC}, \text{AR}\}} \left[1 - \frac{TP_c + 1}{TP_c + \alpha_c FP_c + \beta_c FN_c + 1} - \frac{1}{|\Omega|} \sum_{i \in \Omega} \alpha_{t,c} (1 - p_{t,i})^{\gamma_c} \log p_{t,i} \right], \quad (3.1)$$

with soft counts $TP_c = \sum_i p_i y_i$, $FP_c = \sum_i p_i (1 - y_i)$, $FN_c = \sum_i (1 - p_i) y_i$ per image; TC: $\alpha = 0.3$, $\beta = 0.7$, focal $\alpha = 0.95$, $\gamma = 5$; AR: $\alpha = \beta = 0.5$, focal $\alpha = 0.85$, $\gamma = 5$. Logits are upsampled bilinearly to 768×1152 before the loss. The multi-scale fusion models add their deep-supervision terms (Eq. 3.4). All models use AdamW (learning rate 10^{-3} , weight decay 10^{-4}), cosine decay to 10^{-5} , 60 epochs, an effective batch size of 8 and bf16 autocast; the checkpoint with the best validation mean foreground IoU (evaluated every second epoch) is kept. Every configuration is trained with three seeds.

From a prompter output to SAM prompts. A binary mask $m_c = \mathbb{I}[\ell_c > 0]$ is split into 8-connected components of at least 20 pixels, as in `PromptMaker`. Seven conversions are compared (Table 3.1). Boxes and points are mapped from the 1152×768 grid to SAM’s 1024×1024 frame (the image is stretched, not padded), and the masks of all prompts of a class are merged by a pixel-wise maximum of the logits, as in `ClimateSAM.assemble_raw_masks`.

Conversion	Prompt(s) per connected component
box	tight bounding box
box (+10%)	box enlarged by 10% of its width and height
points	5 positive points in the eroded component + its centroid, 5 negative points in a ring around it
box + points	both of the above
mask logits	one dense prompt per class: ℓ_c resized to 256×256 , clamped to $[-20, 20]$ (hard masks: ± 10)
box + mask	every box together with the dense prompt of its class
hybrid	TC: box + mask logits; AR: mask logits only

Table 3.1: Conversions of a prompter output into SAM prompts. The hybrid conversion follows from the prompt analysis in Section 4.3.1: dense prompts carry the shape of an AR, but a TC covers only a few cells of the prompt encoder’s 64×64 grid and needs a box.

Metrics. As in the other thesis tables, IoU is computed per class on counts summed over all $N = 61$ test images,

$$\text{IoU}_c = \frac{\sum_n TP_{c,n}}{\sum_n (TP_{c,n} + FP_{c,n} + FN_{c,n})}, \quad \text{Mean FG IoU} = \frac{1}{2} (\text{IoU}_{\text{TC}} + \text{IoU}_{\text{AR}}), \quad (3.2)$$

where a pixel predicted as both classes counts as AR. Three further analyses are used:

- **Object level.** A ground-truth object (8-connected, ≥ 20 px) is *found* if any predicted pixel overlaps it; a predicted object is *correct* if it overlaps any ground-truth object. Recall = found / ground-truth objects, precision = correct / predicted objects.
- **Error decomposition.** The IoU of a prediction is recomputed after removing one error type at a time: deleting predicted objects that touch no ground truth (*no false objects*), adding the missed ground-truth objects (*add missed objects*), both (*perfect detection*), or replacing every detected ground-truth object by its exact shape while false objects are kept (*perfect shape*). The gain of each variant measures the IoU lost to that error type.
- **Paired bootstrap.** Two outputs A and B are compared by resampling the test images with replacement (2000 times), recomputing both dataset-level IoUs on each resample, and reporting the 2.5 and 97.5 percentiles of $\text{IoU}_B - \text{IoU}_A$ as a 95% confidence interval. For learned prompters the per-image counts of the three seeds are pooled, so the interval reflects the sampling of test images, which dominates the variance (seed standard deviations are 0.001–0.01).

3.7 Mask-Prompt Generator

The multi-scale fusion generator (Figure 3.9) upsamples all twelve ViT block outputs to 1024×1024 with nearest-neighbour interpolation before it predicts. The mask-prompt generator (MPG) proposed here starts from three observations: (1) the encoder describes the image on a 64×64 grid, so upsampling its features before the prediction adds computation but no information; (2) the only output resolution SAM needs is its dense-prompt resolution, 256×256 ; (3) ARs are long, thin structures, so the prediction needs a large receptive field.

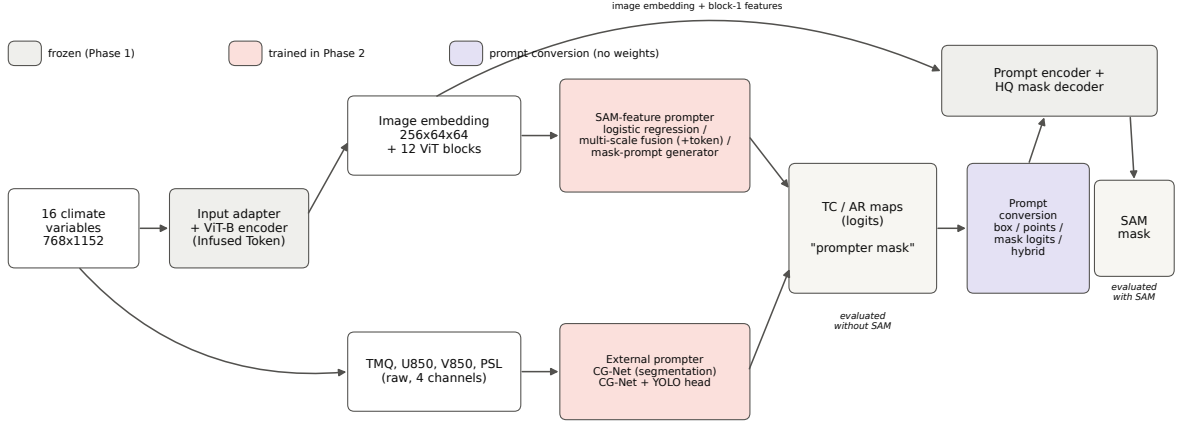


Figure 3.1: Benchmark pipeline for automatic prompting. Every prompter produces TC and AR logit maps (the *prompter mask*), which are evaluated directly and after conversion into prompts for the frozen prompt encoder and HQ decoder. Grey: frozen Phase-1 components; red: trained in Phase 2; violet: prompt conversion (no weights).

MPG projects the neck embedding E and the concatenated block outputs $[F_6; F_{12}]$ to 128 channels each by 1×1 convolutions, adds them, and applies three ConvNeXt blocks at 64×64 with dilations 1, 2 and 4 (depth-wise 3×3 convolution, LayerNorm, 1×1 expansion by 4, GELU, 1×1 projection, residual connection). Two transposed convolutions upsample to 256×256 , and a 3×3 convolution predicts the two logit maps (Figure 3.2). The bias of the last layer is initialised to the logit of a 5% foreground prior, $b = -\log \frac{1-\pi}{\pi}$ with $\pi = 0.05$, so that the initial prompts are nearly empty instead of random. The model has 0.67 M parameters and needs 5.7 GFLOPs per image, compared with 1.41 M parameters and 702 GFLOPs for the multi-scale fusion generator.

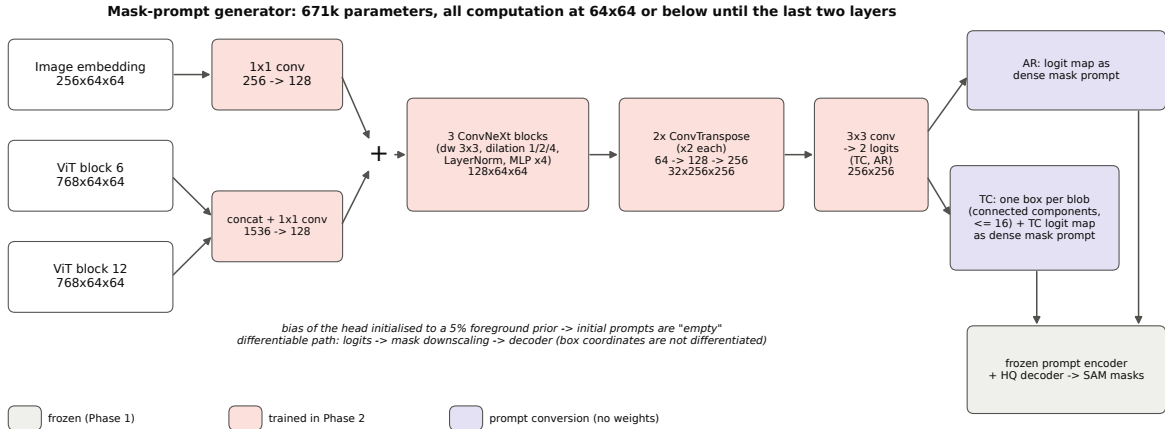


Figure 3.2: Mask-prompt generator and the hybrid conversion of its logit maps into SAM prompts.

Training through SAM. The hybrid conversion is differentiable with respect to the logit maps: the AR map enters the prompt encoder’s mask-downscaling convolutions directly, and the TC map does so together with (non-differentiable) boxes around its connected components.

MPG can therefore also be trained with a loss on SAM’s output,

$$\mathcal{L} = \lambda \mathcal{L}_{\text{seg}}(\ell, y) + \mathcal{L}_{\text{seg}}(\text{SAM}(\text{hybrid}(\ell)), y), \quad (3.3)$$

either from scratch (*end-to-end*, $\lambda = 0.5$) or starting from the model trained with $\mathcal{L}_{\text{seg}}$ alone for 30 further epochs at learning rate 10^{-4} (*two-stage*, $\lambda = 1$). The frozen decoder acts as a fixed differentiable function, so the gradient reaches the prompter (Figure 3.3).

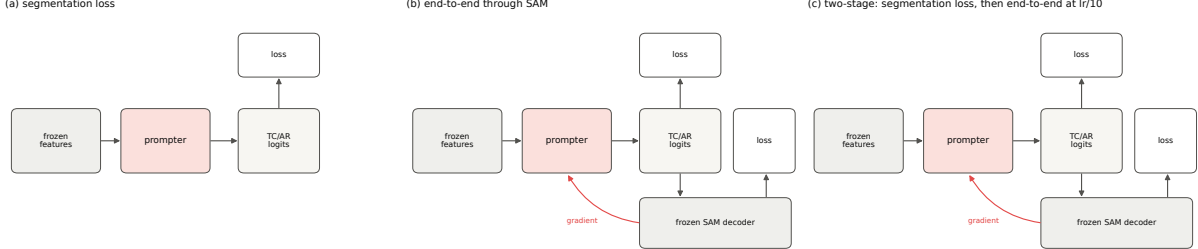


Figure 3.3: The three training modes of the mask-prompt generator: segmentation loss only, end-to-end through the frozen SAM decoder, and two-stage.

3.8 Adapting the Decoder to Generated Prompts

The Phase-1 decoder was trained only with ground-truth prompts. Two ways of exposing it to realistic prompts are tested. In both, the trainable parts of the HQ decoder (`hf_mlp_ar/tc`, `compress_vit_feat`, `embedding_encoder`, `embedding_maskfeature`; 1.21 M parameters) are fine-tuned with the loss of Eq. (3.1) on the union of the prompted masks of a class against the complete ground truth of that class, while encoder and prompt encoder stay frozen.

Adaptation to a prompter. The decoder is trained on the prompts of a fixed prompter (30 epochs, learning rate $3 \cdot 10^{-4}$). A prompter is much better on the images it was trained on than on new ones, so in-sample prompts are unrealistically clean. The *out-of-fold* variant avoids this: five copies of MPG are trained, each without one fifth of the training images, and every training image receives its prompts from the copy that did not see it.

Prompt-robust decoder. The decoder is trained on corrupted ground-truth prompts mixed with out-of-fold MPG prompts (probability 0.5 each; 25 epochs, learning rate $3 \cdot 10^{-4}$). The corruptions mimic the errors of real prompters. Every object is left without a prompt with probability 0.15; every box side is moved by $U(-0.15, 0.3)$ times the box size; Binomial(3, 0.25) false boxes of realistic size are added per class; dense prompts are eroded or dilated with an elliptical kernel of up to 5×5 px and, with probability 0.3, receive a false blob (a real object moved to a random position). The target of a box is the *complete* ground-truth object(s) it touches, and an empty mask for a box that touches no object. Unlike in Phase 1, the decoder is thus trained to output nothing for a false prompt and to extend or shrink a prompt towards the object. Two prompt modes are trained (box, hybrid), and a control variant is trained on corrupted ground-truth prompts only.

Chapter 4

Results

4.3 Automatic Prompt Generator for SAM

Phase 1 showed that the adapted SAM segments TCs and ARs well when it receives ground-truth prompts. An operational system must obtain its prompts automatically, from a *prompter*. This section asks whether such a pipeline (prompter $\rightarrow$ prompts $\rightarrow$ SAM) outperforms the prompter alone, and which prompter and prompt format work best. It is organised along five research questions:

- RQ1** How well can the frozen Phase-1 SAM segment TCs and ARs if the prompts are perfect, and how sensitive is it to realistic prompt errors? (Section 4.3.1)
- RQ2** Does SAM prompted by CG-Net improve on CG-Net, as reported in the earlier version of Table 4.13? (Section 4.3.2)
- RQ3** Which prompter produces the best masks and prompts under identical conditions, and at what cost? (Section 4.3.3)
- RQ4** Does SAM add value to any prompter, and can this be achieved by training the prompter through SAM or by adapting the decoder? (Sections 4.3.3 and 4.3.5)
- RQ5** Where exactly do the automatic systems lose IoU? (Section 4.3.4)

All experiments follow the protocol of Section 3.6: one frozen Phase-1 model (Infused Token, MLP ratio 1.0), checkpoint selection on 40 held-out training images, three seeds per learned prompter, and evaluation on the 61 test images.

4.3.1 Prompt Analysis

Before any prompter is trained, prompts built from the ground truth show what the frozen SAM can achieve and how it reacts to the errors a real prompter makes.

Prompt types. With perfect prompts the tight box is the best sparse prompt (TC 0.723 / AR 0.625), points are the weakest (0.527 / 0.577), and adding points to a box *lowers* the IoU (0.583 / 0.604), as in Table 4.12 (Figure 4.1). Dense prompts are better, because they carry the shape of the object: mask logits alone reach AR 0.920, and the hybrid prompt (TC: box + mask, AR: mask) reaches TC 0.872 / AR 0.920. These numbers are upper bounds, since a ground-truth mask prompt already contains the answer, but they show that the adapted encoder and decoder are not the bottleneck of the automatic pipeline.

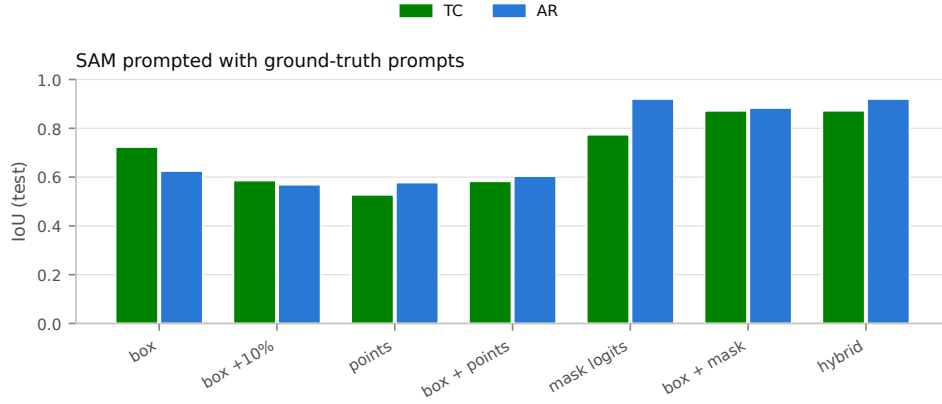


Figure 4.1: SAM prompted with prompts built from the ground truth (test set, primary checkpoint).

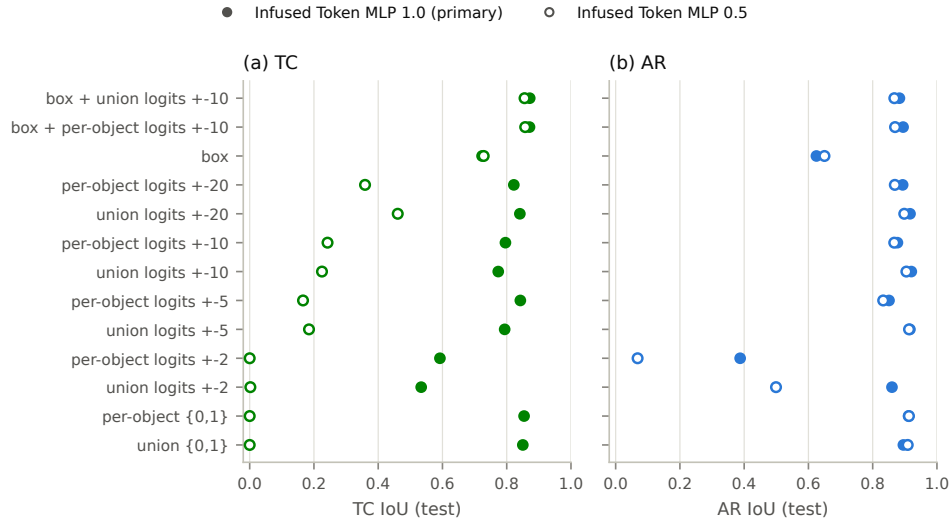


Figure 4.2: How the decoder reads dense mask prompts built from the ground truth: $\{0,1\}$ maps versus logit maps of scale $\pm a$, one prompt per class (union) versus one per object, with and without a box. Filled markers: primary checkpoint (MLP ratio 1.0); open markers: MLP ratio 0.5.

Format of dense prompts. Whether the decoder uses a dense prompt for TCs depends on the checkpoint (Figure 4.2). On the primary checkpoint both $\{0, 1\}$ and logit masks work (TC 0.85, AR 0.90); on the MLP-0.5 checkpoint a $\{0, 1\}$ mask yields no TC at all (0.000) and even ± 20 logits reach only TC 0.46, while ARs are unaffected. The prompt encoder reduces the 256×256 mask to 64×64 , where a cyclone covers only a handful of cells, and whether the decoder learned to use such weak dense evidence is a by-product of its Phase-1 training. A box together with the mask is robust on both checkpoints (TC 0.86–0.87); this motivates the hybrid conversion. One prompt per class works as well as one per object, so a single decoder call per class and image suffices for dense prompts.

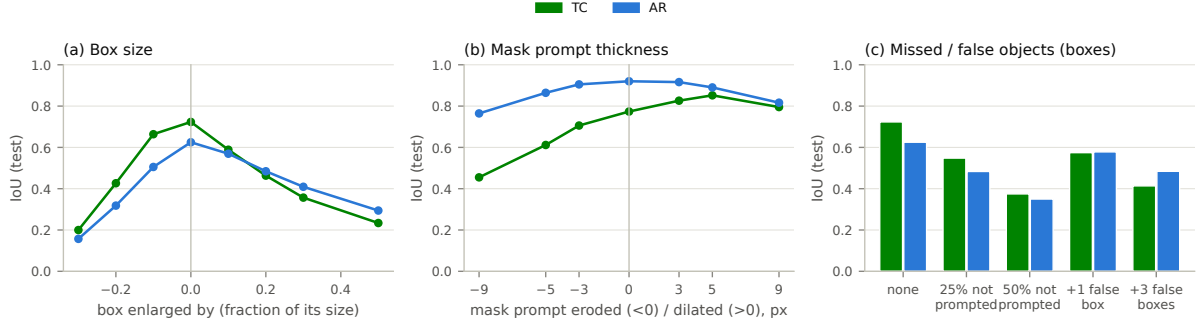


Figure 4.3: SAM with deliberately corrupted ground-truth prompts (primary checkpoint): (a) box size, (b) thickness of a dense mask prompt, (c) missed and false objects.

Ground-truth prompt	MLP 1.0	MLP 0.5	adapted to MPG hybrid prompts (OOF)	adapted to MPG boxes (OOF)	prompt-robust, box	prompt-robust, box (seed 1)	prompt-robust, hybrid	prompt-robust, hybrid (seed 1)	prompt-robust, hybrid, corrupted GT only (selected epoch 0)	prompt-robust, hybrid, corrupted GT only (last epoch)
tight box	0.72 / 0.62	0.73 / 0.65	0.70 / 0.59	0.68 / 0.58	0.68 / 0.59	0.72 / 0.46	0.70 / 0.39	0.72 / 0.62	0.71 / 0.32	0.71 / 0.32
box enlarged 20%	0.46 / 0.48	0.44 / 0.40	0.47 / 0.39	0.50 / 0.52	0.61 / 0.52	0.54 / 0.48	0.61 / 0.44	0.46 / 0.48	0.62 / 0.32	0.62 / 0.32
box shrunk 20%	0.43 / 0.32	0.43 / 0.31	0.48 / 0.29	0.46 / 0.37	0.38 / 0.34	0.43 / 0.36	0.43 / 0.29	0.42 / 0.18	0.43 / 0.32	0.43 / 0.16
box, 25% of objects not prompted	0.55 / 0.48	0.57 / 0.50	0.54 / 0.44	0.50 / 0.46	0.50 / 0.45	0.50 / 0.46	0.54 / 0.34	0.52 / 0.30	0.55 / 0.48	0.52 / 0.24
box + 1 false box per class	0.57 / 0.58	0.58 / 0.58	0.56 / 0.55	0.64 / 0.58	0.66 / 0.56	0.65 / 0.56	0.60 / 0.45	0.62 / 0.39	0.57 / 0.58	0.64 / 0.32
box + 3 false boxes per class	0.41 / 0.48	0.41 / 0.47	0.41 / 0.47	0.60 / 0.54	0.63 / 0.50	0.63 / 0.50	0.46 / 0.41	0.54 / 0.36	0.41 / 0.48	0.55 / 0.31
mask logits ± 10	0.77 / 0.62	0.76 / 0.60	0.72 / 0.58	0.62 / 0.58	0.57 / 0.50	0.54 / 0.50	0.60 / 0.52	0.77 / 0.62	0.60 / 0.52	0.60 / 0.52
mask eroded 5 px	0.61 / 0.36	0.31 / 0.36	0.57 / 0.32	0.48 / 0.30	0.09 / 0.29	0.39 / 0.34	0.30 / 0.36	0.61 / 0.36	0.00 / 0.34	0.00 / 0.34
mask dilated 5 px	0.85 / 0.89	0.87 / 0.88	0.82 / 0.80	0.76 / 0.80	0.71 / 0.80	0.72 / 0.80	0.72 / 0.80	0.80 / 0.80	0.85 / 0.89	0.80 / 0.80
box + mask logits ± 10	0.87 / 0.88	0.86 / 0.87	0.87 / 0.83	0.84 / 0.77	0.82 / 0.86	0.84 / 0.86	0.88 / 0.83	0.86 / 0.87	0.87 / 0.88	0.88 / 0.88

Table 4.1: SAM prompted with prompts built from the ground truth, with controlled errors (TC IoU / AR IoU, 61 test images). Columns: the frozen Phase-1 decoders of both checkpoints and the decoders fine-tuned on generated or corrupted prompts (Section on decoder adaptation).

Sensitivity to prompt errors. Figure 4.3 and Table 4.1 quantify how SAM reacts to the errors a prompter makes:

- *Box size.* The IoU peaks sharply at the tight box. Enlarging all boxes by 20% costs 0.26 TC and 0.14 AR IoU; shrinking them by 20% costs 0.30 and 0.31. The decoder takes the extent of the box literally.
- *Missed and false objects.* An object without a prompt is never recovered (25% of the objects without a prompt: TC 0.723 $\rightarrow$ 0.548, AR 0.625 $\rightarrow$ 0.483), and one random false box per class costs 0.149 TC and 0.047 AR IoU: the decoder segments *something* inside every box it receives.
- *Dense prompts are forgiving.* Eroding the mask by 3 px costs only 0.07 TC / 0.02 AR, and dilating it by 3–5 px even *improves* TC (0.774 $\rightarrow$ 0.852), because the $4\times$ downsampling in the prompt encoder otherwise weakens small objects.

Answer to RQ1. With correct prompts the Phase-1 SAM is an excellent segmenter (hybrid prompts: TC 0.87 / AR 0.92). It does not, however, correct its prompts: missed objects stay missed, false prompts become false objects, and a box error of 10–20% already costs 0.1–0.3 IoU.

4.3.2 SAM with CG-Net as Prompter

Re-evaluation of Table 4.13. Every configuration of the earlier Table 4.13 was re-run with the thesis pipeline itself (the fine-tuned CG-Net, `PromptMaker`, `ClimateSAM.infer` and `forward`) on the 61 test images. Two image paths were compared: the path of `test_prompt_effect.py`, which calls `ClimateSAM.set_infer_img()` and thereby skips the learned input adapter (it feeds the first three raw channels to the encoder), and the training path `encode_images()`, which applies it.

Prompt type	Pos.	Neg.	Enlarge	IoU AR	IoU TC	IoU AR (orig. script)	IoU TC (orig. script)
CG-Net alone (baseline)				0.3817	0.3494		
bbox	0	0	0.0	0.3690	0.3414	0.3630	0.3390
mask	0	0	0.0	0.3758	0.3343	0.3568	0.3251
bbox	0	0	-0.1	0.3267	0.3327	0.3294	0.3315
point+bbox	15	10	0.0	0.3305	0.3252	0.3115	0.3232
point+bbox	10	5	0.0	0.3427	0.3235	0.3254	0.3181
point	5	10	0.0	0.3373	0.3178	0.3236	0.3139
point	1	3	0.0	0.2713	0.3171	0.2774	0.3138
point	5	5	0.0	0.3375	0.3128	0.3259	0.3093
bbox	0	0	0.1	0.3582	0.3073	0.3436	0.3062
point	2	2	0.0	0.3180	0.3048	0.3148	0.2986
point	1	2	0.0	0.2866	0.3038	0.2878	0.3023
point	1	1	0.0	0.3086	0.2964	0.3049	0.2946
point	10	10	0.0	0.3262	0.2923	0.3031	0.2847
bbox	0	0	0.2	0.3334	0.2703	0.3110	0.2706
bbox	0	0	-0.2	0.2288	0.2502	0.2357	0.2506
point+bbox+mask	20	10	0.0	0.3313	0.2396	0.3212	0.2184
bbox	0	0	0.3	0.2991	0.2358	0.2717	0.2376
bbox	0	0	0.5	0.2372	0.1822	0.2042	0.1837
point+bbox+mask	10	5	0.0	0.3471	0.1357	0.3390	0.1222

Table 4.2: SAM with CG-Net as prompter, re-run on the 61 test images (Infused Token MLP 1.0, Phase-1 checkpoint of Table 4.12). “orig. script”: image passed through `ClimateSAM.set_infer_img`, which skips the learned input adapter (`test_prompt_effect.py`); other columns: input adapter applied as in training. Ordered by TC IoU.

The re-run of the original script with the checkpoint of the last recorded run reproduces that run exactly (box prompts: TC 0.3404 / AR 0.3758 versus 0.3409 / 0.3758 in the stored result file), so the pipeline is faithfully reproduced. The SAM rows of the earlier Table 4.13, however, are 0.07–0.12 IoU higher than every re-run and could not be traced to any stored result; they most likely stem from an earlier state of the code. Table 4.2 and Figure 4.4 give the corrected values. The ranking of the prompt types is confirmed (tight boxes > boxes + points > points; enlarging boxes hurts monotonically), but **SAM prompted by CG-Net is never better than CG-Net itself**: the best configuration, tight boxes, reaches TC 0.341 / AR 0.369 against 0.349 / 0.382 for CG-Net alone. The skipped input adapter is a genuine bug but changes the IoU by only 0.00–0.03. The “mask” row depends on the checkpoint, as explained in Section 4.3.1.

CG-Net with a YOLO box head. The thesis also attached a YOLO-style detection head to CG-Net that predicts boxes directly. Its boxes are the weakest prompts of the study (Figure 4.5; SAM: TC 0.19–0.22, AR 0.22–0.29 for the three stored checkpoints and all objectness thresholds), far below boxes derived from CG-Net’s segmentation (0.34 / 0.37). Directly regressed boxes are too loose for thin, curved ARs, and the prompt analysis showed that a box error of 20% alone costs 0.14–0.26 IoU.

CG-Net trained under the protocol. The fine-tuned CG-Net checkpoint used above was trained outside the protocol of Section 3.6, on all 398 training images. For a fair comparison with

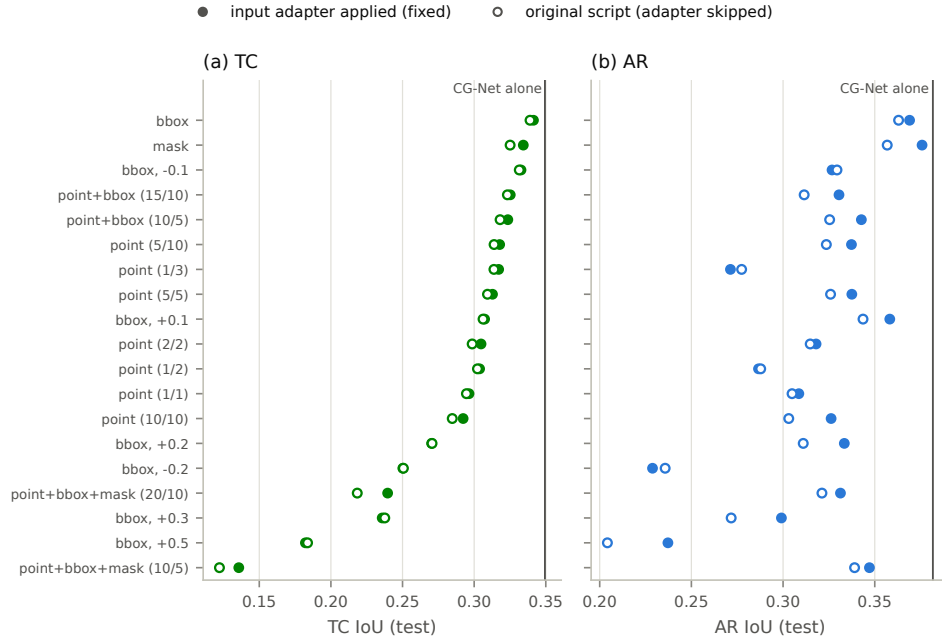


Figure 4.4: SAM with CG-Net as prompt, all configurations of Table 4.13 (primary checkpoint). Filled: input adapter applied; open: original script. Vertical line: CG-Net alone.

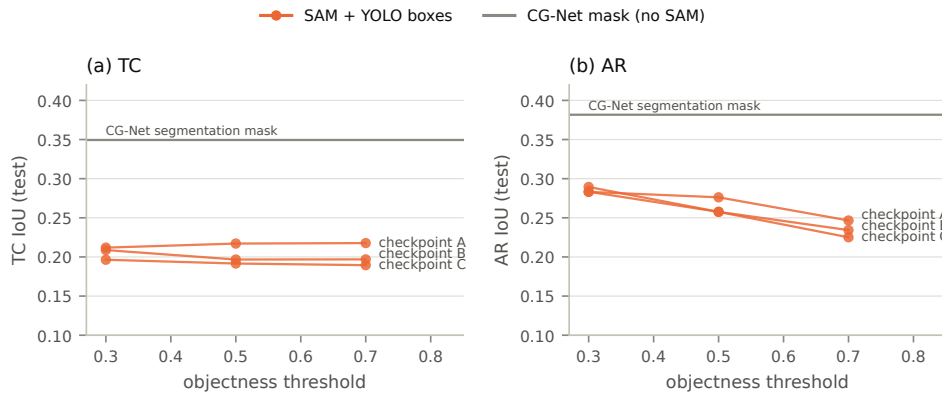


Figure 4.5: SAM prompted by the boxes of the CG-Net YOLO head (three stored checkpoints A–C) as a function of the objectness threshold. Grey line: CG-Net’s own segmentation.

the learned SAM-feature prompters, CG-Net was re-trained on the 358 training images with its original Jaccard loss and the same validation-based checkpoint selection, once starting from the official ClimateNet weights (two seeds, learning rate 10^{-4}) and once from random initialisation (three seeds, learning rate 10^{-3}). Both are clearly better than the stored checkpoint: TC 0.350 ± 0.005 / AR 0.399 ± 0.005 (mean FG IoU 0.375) from the official weights and TC 0.343 ± 0.007 / AR 0.394 ± 0.008 (0.369) from scratch, against 0.349 / 0.382 (0.366) for the stored checkpoint (Table 4.3). A carefully trained CG-Net on the four raw fields is therefore a much stronger baseline than the earlier Table 4.13 suggested, and it is used as the reference prompter in Section 4.3.3. SAM does not improve these masks either: with hybrid prompts the difference is -0.001 mean FG IoU for both variants, with boxes -0.007 to -0.009 .

Answer to RQ2. The ranking of prompt types in Table 4.13 holds, its central claim does not: with correct numbers, SAM prompted by CG-Net is slightly worse than CG-Net for every prompt type.

4.3.3 Generator from SAM Inputs

The prompters of this section read the frozen encoder’s features instead of the raw climate fields: the linear probe (logistic regression) on the first and on the last ViT block, the multi-scale fusion generator (MSF, Figure 3.9) and its token-gated variants, and the mask-prompt generator (MPG, Section 3.7). The original multi-scale generators predicted three channels, of which the loss supervised only TC and AR as independent sigmoids while the prediction was the three-channel argmax, so an untrained background channel competed with TC and AR; here all prompters use two sigmoid channels thresholded at zero. The original logistic-regression prompter read the *first* ViT block; it is evaluated as is and on the last block.

Prompters. Table 4.3 and Figure 4.6 compare all prompters. The mask-prompt generator gives the best prompter mask (TC 0.344 ± 0.003 , AR 0.413 ± 0.001). The paired bootstrap (Table 4.4) locates its advantage over the fine-tuned CG-Net entirely in the ARs (AR $+0.031$, 95% CI $[+0.020, +0.043]$; TC -0.005 , $[-0.030, +0.020]$); it is also better than MSF ($+0.011$ mean FG IoU, $[+0.003, +0.019]$) and than the linear probe on the last block ($+0.036$, $[+0.025, +0.046]$). The token gate *lowers* MSF slightly but significantly (-0.006 , $[-0.009, -0.003]$): its inputs are the fixed HQ tokens of the frozen decoder, so the gate is a learned per-class channel weighting that carries no image-dependent information. Label smoothing, which helped in Phase 1, does not help the prompter (-0.004 , $[-0.007, -0.000]$). A linear probe on the last ViT block (1.5k parameters) already reaches TC 0.301 / AR 0.385, close to the fine-tuned CG-Net, while the same probe on the first block, the original configuration, reaches only 0.195 / 0.319: the late features of the adapted encoder are almost linearly separable with respect to TC and AR, the early ones are not.

Comparison with the re-trained CG-Net. Against CG-Net trained under the same protocol, the advantage of MPG shrinks to the ARs (Table A.2): AR $+0.018$ (95% CI $[+0.009, +0.027]$) over CG-Net from scratch and $+0.013$ ($[+0.003, +0.024]$) over CG-Net from the official weights, while TC is equal ($+0.001$ and -0.006 , both intervals spanning ± 0.02). The mean FG IoU difference ($+0.010$ and $+0.004$) is not significant. The prompters built on the adapted SAM encoder thus match a well-trained dedicated network on the raw fields and are slightly better for ARs, but they do not surpass it by a large margin.

Adding the raw fields to MPG. Since CG-Net reaches nearly the same quality from only four raw fields, a variant of MPG additionally receives these fields (average-pooled to 64×64 before the ConvNeXt blocks and resized to 256×256 before the head, both branches initialised

Prompter	Output	TC IoU	AR IoU	BG IoU	Mean IoU	Mean FG IoU
Ground truth (oracle)	Prompter mask (no SAM)	1.000	1.000	1.000	1.000	1.000
	SAM + box	0.723	0.625	0.971	0.773	0.674
	SAM + hybrid	0.872	0.920	0.994	0.929	0.896
CG-Net (official weights)	Prompter mask (no SAM)	0.327	0.333	0.926	0.529	0.330
	SAM + box	0.318	0.342	0.931	0.530	0.330
	SAM + hybrid	0.327	0.339	0.929	0.532	0.333
CG-Net (fine-tuned)	Prompter mask (no SAM)	0.349	0.382	0.940	0.557	0.366
	SAM + box	0.341	0.369	0.939	0.550	0.355
	SAM + hybrid	0.349	0.384	0.941	0.558	0.367
CG-Net (re-trained on 358 images, official init.)	Prompter mask (no SAM)	0.350±0.005	0.399±0.005	0.945±0.001	0.565±0.003	0.375±0.005
	SAM + box	0.345±0.003	0.387±0.006	0.943±0.001	0.558±0.003	0.366±0.004
	SAM + hybrid	0.348±0.007	0.400±0.005	0.946±0.001	0.564±0.004	0.374±0.006
CG-Net (trained from scratch on 358 images)	Prompter mask (no SAM)	0.343±0.007	0.394±0.008	0.946±0.004	0.561±0.003	0.369±0.006
	SAM + box	0.343±0.006	0.380±0.002	0.942±0.004	0.555±0.002	0.362±0.003
	SAM + hybrid	0.339±0.008	0.395±0.008	0.946±0.003	0.560±0.003	0.367±0.006
Logistic regression, ViT block 1	Prompter mask (no SAM)	0.195±0.001	0.319±0.001	0.927±0.004	0.480±0.002	0.257±0.001
	SAM + box	0.200±0.002	0.321±0.003	0.925±0.004	0.482±0.003	0.260±0.002
	SAM + hybrid	0.213±0.000	0.316±0.002	0.939±0.001	0.489±0.001	0.264±0.001
Logistic regression, ViT block 12	Prompter mask (no SAM)	0.301±0.004	0.385±0.001	0.938±0.001	0.541±0.002	0.343±0.002
	SAM + box	0.291±0.001	0.362±0.001	0.932±0.001	0.528±0.000	0.326±0.000
	SAM + hybrid	0.304±0.004	0.373±0.002	0.946±0.000	0.541±0.002	0.338±0.002
Multi-scale fusion	Prompter mask (no SAM)	0.333±0.002	0.403±0.004	0.944±0.003	0.560±0.001	0.368±0.003
	SAM + box	0.339±0.001	0.385±0.002	0.941±0.003	0.555±0.001	0.362±0.001
	SAM + hybrid	0.330±0.002	0.388±0.010	0.949±0.001	0.556±0.003	0.359±0.005
Multi-scale fusion + token gate	Prompter mask (no SAM)	0.324±0.001	0.399±0.003	0.944±0.002	0.556±0.001	0.362±0.001
	SAM + box	0.336±0.002	0.382±0.003	0.940±0.002	0.553±0.001	0.359±0.002
	SAM + hybrid	0.329±0.004	0.387±0.006	0.949±0.001	0.555±0.003	0.358±0.005
Multi-scale fusion + token gate, CG blocks	Prompter mask (no SAM)	0.324±0.017	0.355±0.022	0.913±0.014	0.531±0.006	0.340±0.003
	SAM + box	0.289±0.010	0.335±0.021	0.913±0.012	0.512±0.008	0.312±0.006
	SAM + hybrid	0.322±0.017	0.391±0.001	0.934±0.002	0.549±0.005	0.356±0.009
Multi-scale fusion + token gate, shared weights	Prompter mask (no SAM)	0.332±0.006	0.402±0.001	0.943±0.002	0.559±0.002	0.367±0.003
	SAM + box	0.338±0.006	0.385±0.002	0.940±0.002	0.554±0.003	0.362±0.003
	SAM + hybrid	0.333±0.004	0.391±0.005	0.949±0.001	0.558±0.002	0.362±0.003
Mask-prompt generator (segmentation loss)	Prompter mask (no SAM)	0.344±0.003	0.413±0.001	0.944±0.000	0.567±0.002	0.379±0.002
	SAM + box	0.343±0.006	0.391±0.001	0.940±0.001	0.558±0.002	0.367±0.003
	SAM + hybrid	0.342±0.004	0.398±0.002	0.949±0.000	0.563±0.002	0.370±0.002
Mask-prompt generator (end-to-end via SAM)	Prompter mask (no SAM)	0.335±0.010	0.378±0.019	0.941±0.006	0.551±0.010	0.356±0.014
	SAM + box	0.336±0.011	0.385±0.005	0.940±0.007	0.554±0.003	0.360±0.005
	SAM + hybrid	0.334±0.009	0.402±0.007	0.946±0.003	0.561±0.004	0.368±0.008
Mask-prompt generator (two-stage)	Prompter mask (no SAM)	0.338±0.005	0.409±0.002	0.938±0.002	0.562±0.000	0.374±0.001
	SAM + box	0.334±0.003	0.386±0.004	0.935±0.002	0.552±0.001	0.360±0.001
	SAM + hybrid	0.337±0.007	0.411±0.002	0.947±0.001	0.565±0.002	0.374±0.004
Mask-prompt generator (segmentation loss, label smoothing)	Prompter mask (no SAM)	0.338±0.009	0.412±0.001	0.943±0.001	0.564±0.003	0.375±0.005
	SAM + box	0.340±0.007	0.388±0.003	0.939±0.001	0.556±0.002	0.364±0.003
	SAM + hybrid	0.335±0.007	0.401±0.002	0.949±0.000	0.562±0.003	0.368±0.005
Mask-prompt generator + raw CG-Net fields	Prompter mask (no SAM)	0.340±0.001	0.409±0.006	0.942±0.004	0.564±0.004	0.375±0.004
	SAM + box	0.340±0.001	0.387±0.008	0.938±0.004	0.555±0.004	0.363±0.004
	SAM + hybrid	0.337±0.003	0.399±0.006	0.949±0.001	0.561±0.002	0.368±0.003
Box head on SAM features (YOLO-style)	Prompter mask (no SAM)	0.094±0.014	0.156±0.016	0.693±0.035	0.315±0.012	0.125±0.001
	SAM + box	0.107±0.017	0.303±0.025	0.884±0.004	0.431±0.004	0.205±0.004
Learned static prompts (4 tokens / class)	SAM + learned static prompts	0.237±0.001	0.324±0.008	0.926±0.005	0.495±0.004	0.280±0.004
Learned static prompts (16 tokens / class)	SAM + learned static prompts	0.248	0.329	0.930	0.502	0.288

Table 4.3: Automatic prompting on the ClimateNet test set (61 images), frozen ClimateSAM (Infused Token, MLP ratio 1.0). Learned prompts: mean±std over 3 seeds, checkpoint selected on 40 held-out training images.

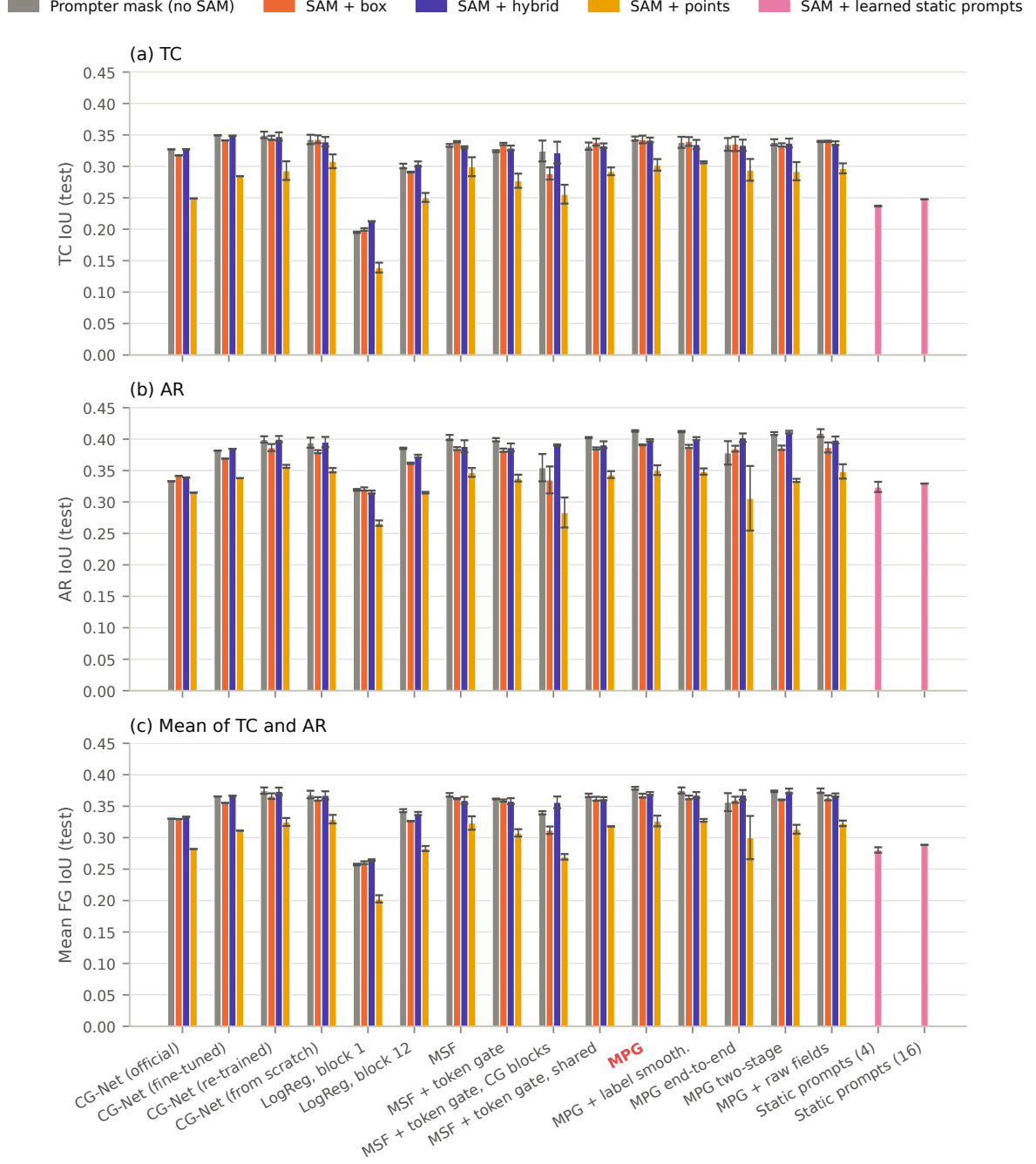


Figure 4.6: Prompter mask (grey) versus SAM prompted by it with boxes (orange), hybrid prompts (violet) and points (yellow); learned static prompts (magenta) have no prompter. Error bars: standard deviation over three seeds.

to zero). It is slightly *worse* than MPG (TC 0.340 ± 0.001 / AR 0.409 ± 0.006 ; -0.004 mean FG IoU, $[-0.008, -0.001]$). The adapted encoder already sees all 16 input channels through the input adapter, so the four fields add no information that the SAM features lack.

Further token-gated variants. Two further multi-scale generators of the thesis were run under the protocol. The variant with context-guided (CG) blocks (1.31 M parameters, 452 GFLOPs) trains slowly (1.5–3.5 hours per seed on cached features, two seeds) and is clearly worse than the plain token gate: TC 0.324 ± 0.017 / AR 0.355 ± 0.022 (mean FG IoU 0.340; -0.023 , 95% CI $[-0.032, -0.015]$), with its best validation epochs early (epochs 2 and 14 of 60). Its AR mask is fragmented (20.5 AR objects per image), which is also the only case in which SAM with hybrid prompts clearly improves on the prompter ($+0.017$, AR $+0.037$) – the result, 0.356, is still below the plain MPG mask. The variant with weights shared across the four block groups (0.39 M parameters, 612 GFLOPs) is slightly better than the token gate ($+0.006$, $[+0.003, +0.009]$; TC 0.332 ± 0.006 / AR 0.402 ± 0.001) but stays below MPG (-0.011 , $[-0.020, -0.004]$).

Box head on SAM features. The thesis also designed a YOLO-style detection head on the image embedding, gated by the decoder’s HQ tokens (0.61 M parameters), whose boxes prompt SAM directly. Trained on ground-truth boxes (two seeds, objectness threshold 0.7 selected on the validation images), it is the weakest image-dependent prompter: SAM with its boxes reaches TC 0.107 ± 0.017 / AR 0.303 ± 0.025 (mean FG IoU 0.205). It finds most objects (TC recall 0.81–0.93), but SAM’s output then contains 9–17 TC objects per image, of which only 17–29% touch a cyclone. As for the CG-Net YOLO head, regressed boxes are too loose and too numerous; every false box becomes a false object (Section 4.3.1).

Comparison (B – A)	Δ TC [95% CI]	Δ AR [95% CI]	Δ mean FG [95% CI]
MPG vs. CG-Net (fine-tuned), masks	-0.005 $[-0.030, +0.020]$	$+0.031$ $[+0.020, +0.043]$	$+0.013$ $[-0.000, +0.027]$
MPG vs. MSF, masks	$+0.011$ $[-0.003, +0.025]$	$+0.010$ $[+0.005, +0.016]$	$+0.011$ $[+0.003, +0.019]$
MPG vs. LogReg block 12, masks	$+0.044$ $[+0.025, +0.062]$	$+0.028$ $[+0.020, +0.036]$	$+0.036$ $[+0.025, +0.046]$
MSF + token gate vs. MSF, masks	-0.009 $[-0.015, -0.003]$	-0.004 $[-0.005, -0.002]$	-0.006 $[-0.009, -0.003]$
MPG: SAM + box vs. its mask	-0.002 $[-0.009, +0.006]$	-0.022 $[-0.028, -0.016]$	-0.012 $[-0.017, -0.007]$
MPG: SAM + hybrid vs. its mask	-0.002 $[-0.007, +0.003]$	-0.015 $[-0.024, -0.006]$	-0.008 $[-0.014, -0.003]$
CG-Net (ft.): SAM + box vs. its mask	-0.008 $[-0.017, +0.001]$	-0.013 $[-0.018, -0.006]$	-0.010 $[-0.016, -0.004]$
MPG end-to-end: SAM + hybrid vs. its mask	-0.001 $[-0.005, +0.003]$	$+0.024$ $[+0.021, +0.028]$	$+0.012$ $[+0.009, +0.015]$
MPG two-stage: SAM + hybrid vs. MPG mask	-0.007 $[-0.013, -0.000]$	-0.002 $[-0.007, +0.003]$	-0.004 $[-0.008, -0.000]$

Table 4.4: Selected paired-bootstrap comparisons on the test set (2000 resamples of the 61 images, seeds pooled). The full list is given in Table A.2.

Added value of SAM. Figure 4.7 answers the central question of Phase 2. Point prompts always lose (-0.03 to -0.08 mean FG IoU). Tight boxes are roughly neutral for TCs (-0.010 to $+0.011$) and lose 0.013–0.024 AR IoU for every prompter with a good AR mask; they are neutral or slightly positive only where the prompter’s AR mask is poor (official CG-Net, linear probe on block 1, end-to-end MPG). The hybrid prompt is the best conversion: within about ± 0.01 of the prompter mask for every prompter, slightly *positive* for prompters with a poor mask (linear probe on block 1: $+0.007$; end-to-end MPG: $+0.012$) and slightly *negative* for the strong ones (MSF -0.009 , MPG -0.008). The best SAM output of the study (two-stage MPG, hybrid prompts, 0.374 mean FG IoU) does not exceed the best prompter mask (MPG, 0.379). The complete comparison of all seven conversions is given in Table A.1.

Training through SAM. Training MPG end-to-end through the frozen decoder makes *its own* mask worse (0.356 versus 0.379) and SAM’s output better than that mask ($+0.012$): the prompter learns to produce logits that SAM turns into good masks. The resulting SAM output

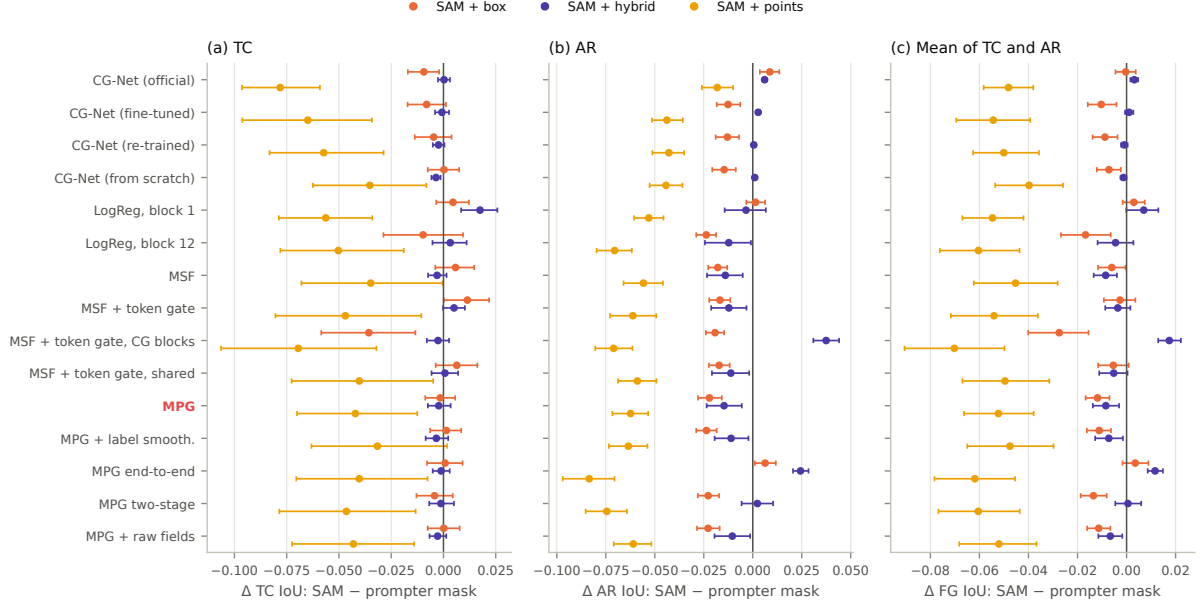


Figure 4.7: Effect of SAM: IoU of SAM’s output minus IoU of the prompt’s own mask, per prompt and prompt type, with paired-bootstrap 95% intervals.

(0.368) nevertheless stays below the plain MPG mask (-0.011 , $[-0.016, -0.005]$), and the end-to-end mask is strongly fragmented (Table 4.6), because a loss on SAM’s output does not penalise speckle that SAM ignores. The two-stage variant keeps the MPG mask quality (0.374) and gives the best SAM output of the study (0.374), again not above the MPG mask (-0.004 , $[-0.008, -0.000]$). Its selected validation epochs are 2–8 of 30: once the prompt has been trained on its own maps, training through SAM barely changes it.

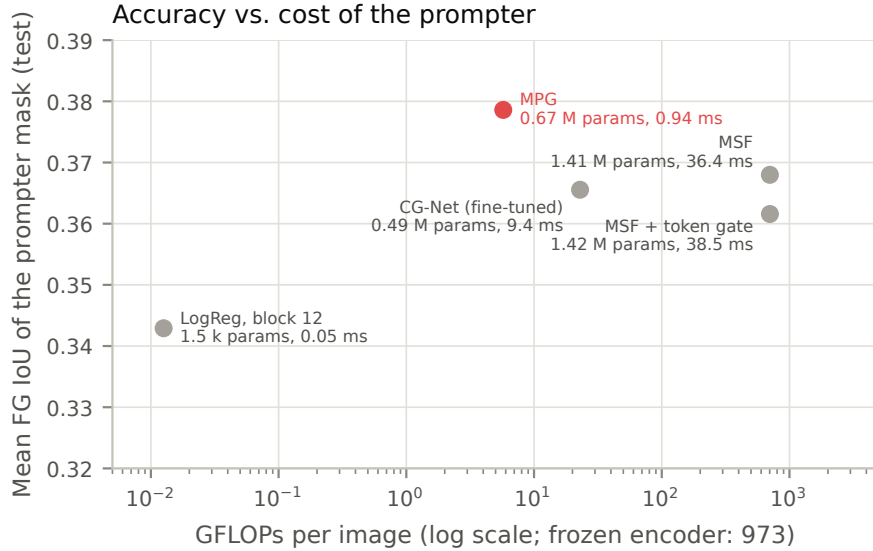


Figure 4.8: Mean FG IoU of the prompt mask versus forward cost per image (log scale).

Cost. MPG is both the most accurate and one of the cheapest prompts (Figure 4.8, Table 4.5): 5.7 GFLOPs and 0.9 ms per image, i.e. $122\times$ fewer FLOPs and $40\times$ lower latency than MSF and $4\times$ fewer FLOPs than CG-Net. Relative to the frozen encoder (973 GFLOPs)

Model	Parameters	GFLOPs / image	Latency (ms, A100, batch 1)
CG-Net	494,232	22.93	9.41
Logistic regression	1,538	0.01	0.05
Multi-scale fusion	1,406,474	702.36	36.36
Multi-scale fusion + token gate	1,415,180	702.89	38.48
Multi-scale fusion + token gate, CG blocks	1,306,724	452.44	62.83
Multi-scale fusion + token gate, shared weights	392,198	612.43	35.38
Mask-prompt generator	671,138	5.74	0.94
Mask-prompt generator + raw fields	686,306	7.14	1.14
(frozen ClimateSAM encoder, for reference)	95,987,456	973.25	123.15
(one SAM decoder call, one box, for reference)	5,269,508	16.80	3.55

Table 4.5: Cost of the prompts on top of the frozen ClimateSAM encoder (forward pass, one image; FLOPs counted with `torch.utils.flop_counter`, latency = median of 20 runs). The encoder and one decoder call are listed for reference.

the prompter adds 0.6%. MSF spends almost all of its computation on 3×3 convolutions at 1024×1024 over features that were upsampled by nearest-neighbour interpolation from 64×64 , i.e. on a resolution that contains no additional information. With cached features a complete MPG training takes about 4 minutes and converges within 10–15 epochs (Figure 4.9); an MSF training takes 46–83 minutes.

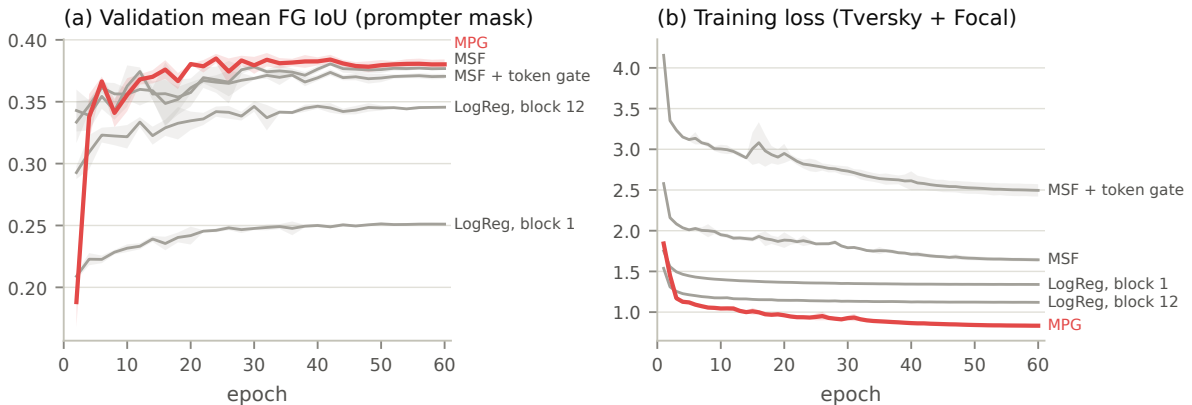


Figure 4.9: Validation mean FG IoU of the prompter mask and training loss (mean $\pm$ std over three seeds). Loss values are not comparable across architectures, because the MSF models add deep-supervision terms.

Answer to RQ3 and RQ4. The best single prompters are the small mask-prompt generator and a CG-Net trained under the same protocol; MPG is better for ARs and far cheaper than the other SAM-feature prompters. SAM prompted by any prompter is at best as good as that prompter, and training the prompter through SAM does not change this.

4.3.4 Error Analysis

Detection. All prompters find more than 90% of the ARs but only about two thirds of the TCs, and 40–60% of their predicted objects touch no ground-truth object (Table 4.6, Figure 4.10). For ARs many of these are small fragments: the prompters predict 12–18 AR objects per image, against 6.8 in the ground truth.

Detection versus delineation. The error decomposition (Figure 4.11, Table A.3) separates the two error sources.

Prompter	TC recall	TC precision	AR recall	AR precision	TC obj./img	AR obj./img
Ground truth (oracle)	1.000	1.000	1.000	1.000	2.9	6.8
CG-Net (official weights)	0.726	0.411	0.935	0.571	5.1	11.1
CG-Net (fine-tuned)	0.682	0.513	0.908	0.567	3.9	11.9
CG-Net (re-trained on 358 images, official init.)	0.687±0.008	0.521±0.044	0.903±0.017	0.657±0.011	3.9	9.8
CG-Net (trained from scratch on 358 images)	0.637±0.015	0.562±0.013	0.914±0.018	0.623±0.054	3.3	10.7
Logistic regression, ViT block 1	0.767±0.006	0.265±0.016	0.967±0.004	0.324±0.018	8.8	25.2
Logistic regression, ViT block 12	0.739±0.012	0.419±0.018	0.966±0.004	0.418±0.010	5.1	17.6
Multi-scale fusion	0.670±0.030	0.529±0.061	0.956±0.001	0.482±0.017	4.0	17.4
Multi-scale fusion + token gate	0.670±0.020	0.448±0.016	0.960±0.005	0.444±0.025	4.7	20.3
Multi-scale fusion + token gate, CG blocks	0.698±0.040	0.492±0.073	0.978±0.007	0.331±0.060	4.2	20.5
Multi-scale fusion + token gate, shared weights	0.680±0.018	0.472±0.018	0.956±0.005	0.459±0.013	4.4	17.7
Mask-prompt generator (segmentation loss)	0.661±0.009	0.563±0.031	0.951±0.001	0.514±0.024	3.5	14.9
Mask-prompt generator (end-to-end via SAM)	0.652±0.012	0.521±0.050	0.963±0.007	0.362±0.107	3.7	90.0
Mask-prompt generator (two-stage)	0.667±0.006	0.546±0.039	0.962±0.003	0.430±0.008	3.7	18.0
Mask-prompt generator (segmentation loss, label smoothing)	0.648±0.015	0.558±0.026	0.953±0.010	0.495±0.007	3.5	15.2
Mask-prompt generator + raw CG-Net fields	0.657±0.009	0.516±0.013	0.950±0.010	0.502±0.054	3.8	14.9

Table 4.6: Object-level detection of the prompter masks: a ground-truth object is found if any predicted pixel overlaps it; a predicted object is correct if it overlaps any ground-truth object. Objects are 8-connected components of at least 20 pixels; with this counting the test set has 2.9 TC and 6.8 AR objects per image.

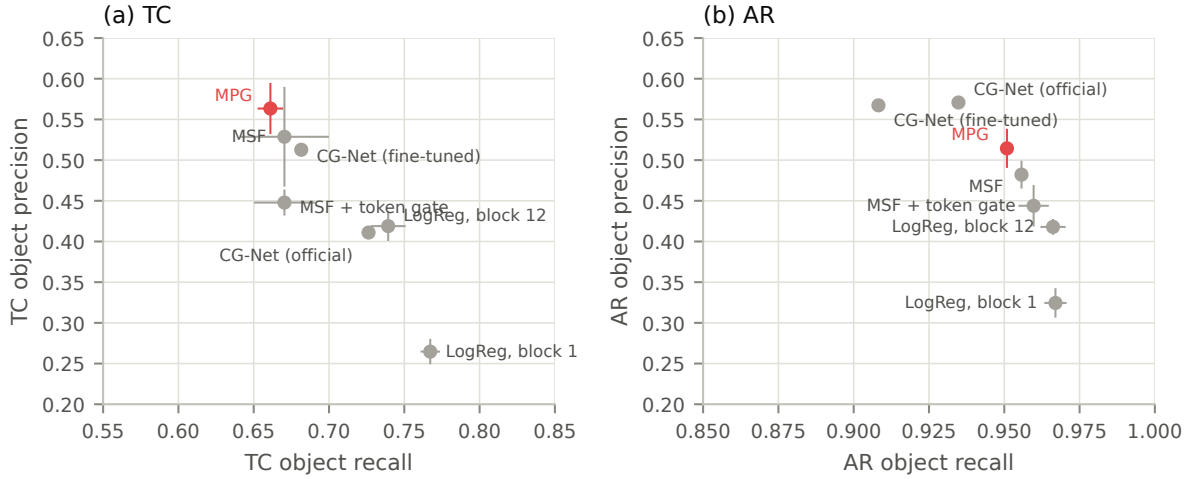


Figure 4.10: Object-level recall and precision of the prompter masks (mean over seeds; bars: standard deviation).

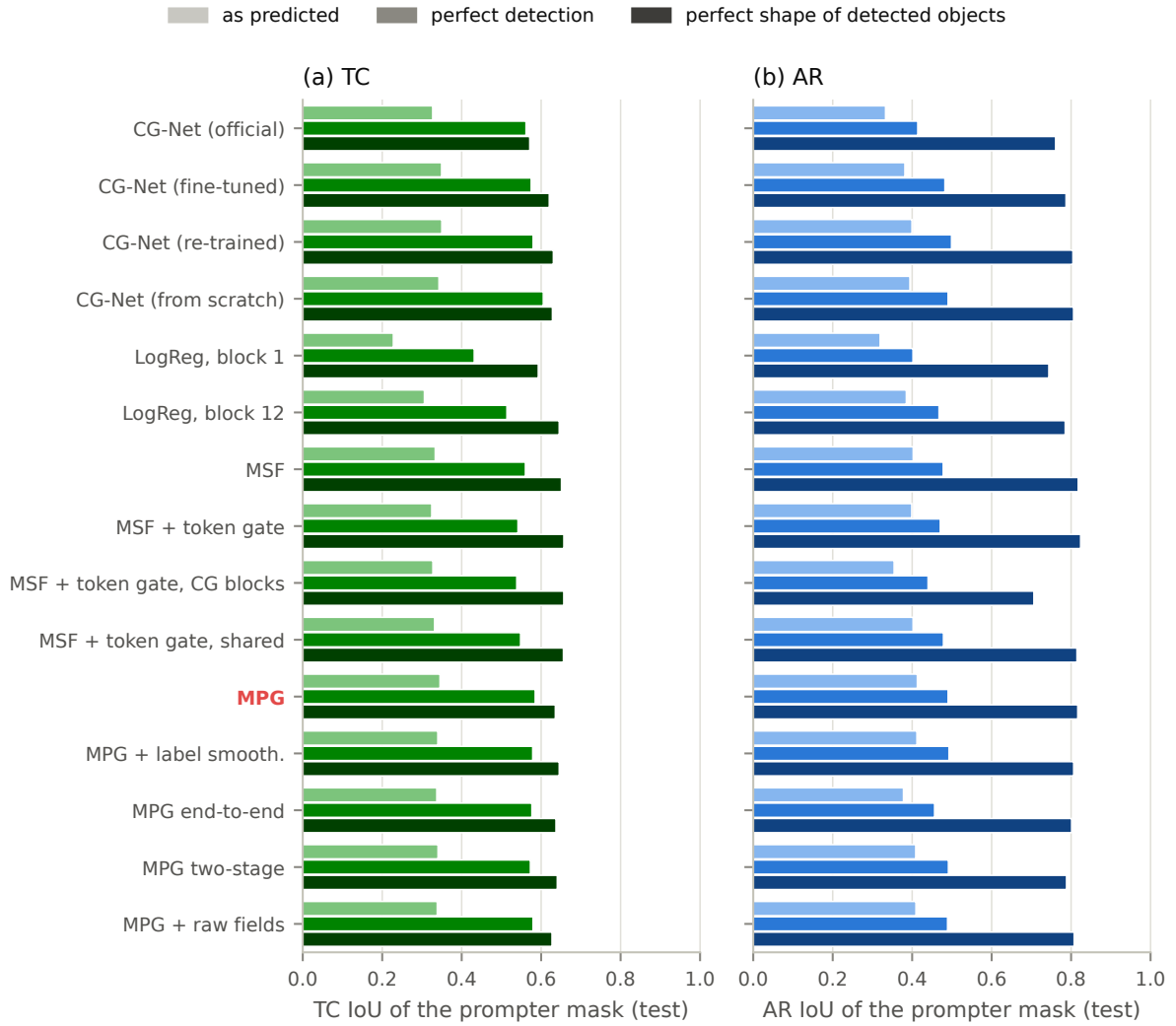


Figure 4.11: IoU of the prompter masks as predicted, with perfect detection (false objects removed, missed objects added), and with the true shape of every detected object (false objects kept).

- *ARs*. For MPG, perfect detection raises the AR IoU only from 0.413 to 0.490, but giving every *detected* river its true shape raises it to 0.817. The rivers are found; what is wrong is their extent, i.e. their length, their width and where a river ends.
- *TCs*. Perfect detection (0.345 $\rightarrow$ 0.585) and perfect shapes of the detected cyclones ($\rightarrow$ 0.636) matter about equally: a third of the cyclones is missed, half of the predicted blobs are false, and the small objects are also delineated imprecisely.

These are exactly the errors that SAM cannot repair (Section 4.3.1): missed objects stay missed, false prompts become false objects, and the extent of an AR is taken from the prompt. This is the mechanism behind the result of Section 4.3.3.

Answer to RQ5. ARs are detected ($> 90\%$) but their extent is wrong; TCs are missed (one third) and falsely detected (about half of the predicted blobs).

4.3.5 Adapting the Decoder to Generated Prompts

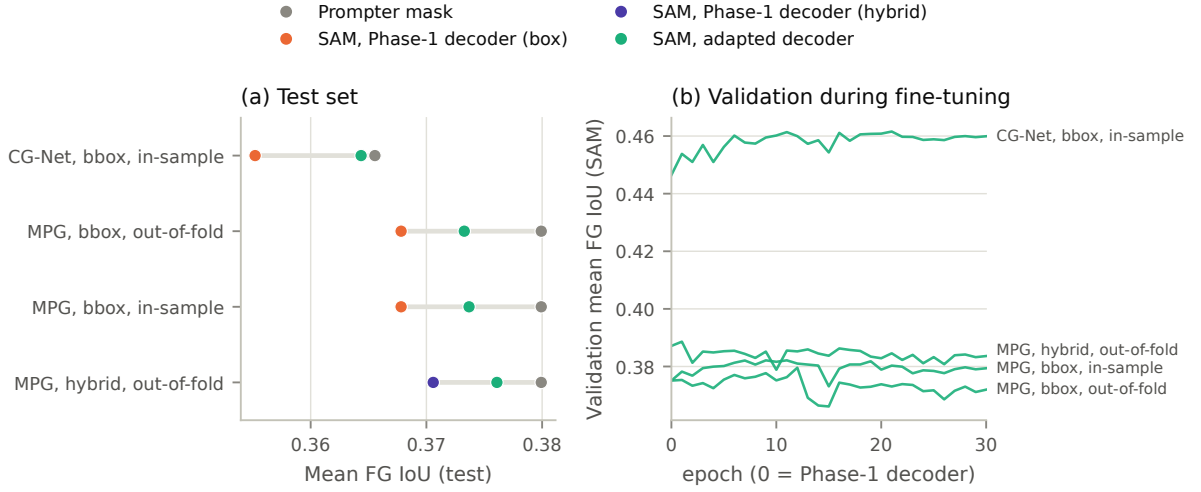


Figure 4.12: (a) Test mean FG IoU of the prompter mask, SAM with the Phase-1 decoder and SAM with the adapted decoder; (b) validation IoU during fine-tuning (epoch 0: Phase-1 decoder; the CG-Net curve is high because CG-Net was trained on these images).

Prompter, prompts, training prompts	Output	TC IoU	AR IoU	Mean FG IoU	TC recall / prec.	AR recall / prec.
CG-Net (fine-tuned), bbox, in-sample	prompter mask	0.349	0.382	0.366		
	SAM, Phase-1 decoder	0.341	0.369	0.355	0.67 / 0.56	0.90 / 0.53
	SAM, adapted decoder	0.345	0.384	0.364	0.64 / 0.58	0.88 / 0.58
Mask-prompt generator, bbox, out-of-fold	prompter mask	0.346	0.414	0.380		
	SAM, Phase-1 decoder	0.344	0.392	0.368	0.65 / 0.59	0.95 / 0.50
	SAM, adapted decoder	0.342	0.404	0.373	0.65 / 0.57	0.93 / 0.62
Mask-prompt generator, bbox, in-sample	prompter mask	0.346	0.414	0.380		
	SAM, Phase-1 decoder	0.344	0.392	0.368	0.65 / 0.59	0.95 / 0.50
	SAM, adapted decoder	0.343	0.404	0.374	0.65 / 0.59	0.95 / 0.55
Mask-prompt generator, hybrid, out-of-fold	prompter mask	0.346	0.414	0.380		
	SAM, Phase-1 decoder	0.344	0.397	0.371	0.65 / 0.59	0.91 / 0.65
	SAM, adapted decoder	0.347	0.405	0.376	0.65 / 0.60	0.94 / 0.64

Table 4.7: Adapting the HQ decoder parts to generated prompts (test set). Phase-1 decoder = the frozen decoder of all other experiments; adapted = best validation epoch of 30.

Adaptation to a prompter. Fine-tuning the decoder on generated prompts (Section 3.8) teaches it something useful: with out-of-fold MPG boxes it rejects part of the false AR prompts (AR object precision $0.50 \rightarrow 0.62$) and it recovers most of what SAM loses relative to the prompter (Figure 4.12, Table 4.7). In no configuration, however, does SAM end up above the prompter mask (best: -0.004 mean FG IoU with out-of-fold hybrid prompts); on the second checkpoint the best configuration ties the prompter ($+0.001$).

Prompt-robust decoder. The adaptation above changes only which prompts the decoder sees. The prompt-robust decoder (Section 3.8) is additionally trained on deliberately corrupted ground-truth prompts whose targets are the complete objects, and empty masks for false boxes. Table 4.8 and Figure 4.13 report each decoder on clean and corrupted ground-truth prompts (as in Table 4.1) and with the prompts of MPG and of the fine-tuned CG-Net.

Decoder	GT box	GT box +20%	GT box + 1 false	GT box + mask	MPG: box	MPG: hybrid	CG-Net: box	CG-Net: hybrid
Phase-1 decoder	0.674	0.474	0.576	0.877	0.367	0.370	0.355	0.367
adapted to MPG hybrid prompts (OOF)	0.644	0.468	0.557	0.851	0.359	0.375	0.342	0.365
adapted to MPG boxes (OOF)	0.634	0.551	0.609	0.803	0.372	0.364	0.366	0.367
prompt-robust, box	0.630	0.559	0.609	0.838	0.371	0.360	0.363	0.368
prompt-robust, box (seed 1)	0.636	0.564	0.608	0.847	0.372	0.364	0.365	0.369
prompt-robust, hybrid	0.592	0.511	0.524	0.856	0.332	0.375	0.320	0.367
prompt-robust, hybrid (seed 1)	0.548	0.527	0.503	0.867	0.317	0.376	0.309	0.369
prompt-robust, hybrid, corrupted GT only (selected: epoch 0)	0.674	0.474	0.576	0.877	0.367	0.370	0.355	0.367
prompt-robust, hybrid, corrupted GT only (last epoch)	0.517	0.468	0.477	0.881	0.291	0.360	0.287	0.367

Table 4.8: Mean FG IoU on the test set of SAM with replaced decoders. GT: prompts built from the ground truth (clean or corrupted, as in the oracle study); MPG: SAM prompted by the mask-prompt generator (mean of three seeds; its own mask reaches 0.379); CG-Net: fine-tuned CG-Net (own mask 0.366).

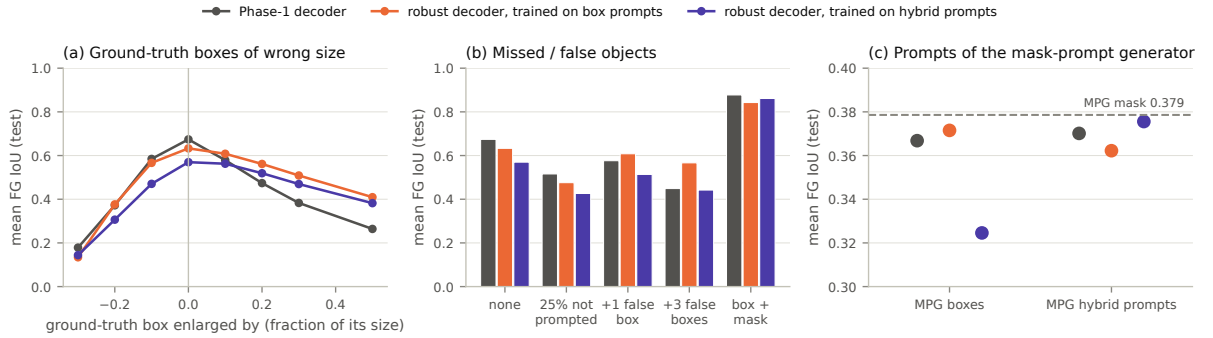


Figure 4.13: Prompt-robust decoders (mean of two seeds) versus the Phase-1 decoder. (a, b) Ground-truth prompts with controlled errors; (c) prompts of the mask-prompt generator (mean of three seeds); dashed: the generator’s own mask.

- *The decoder does become more robust to box errors.* Trained in box mode, it loses less on enlarged boxes (ground-truth boxes enlarged by 20%: mean FG IoU $0.474 \rightarrow 0.559\text{--}0.564$) and on false boxes ($0.576 \rightarrow 0.608\text{--}0.609$), at the price of tight boxes ($0.674 \rightarrow 0.630\text{--}0.636$): it no longer takes the box extent literally.
- *This robustness hardly transfers to real prompters.* SAM with MPG boxes improves by only $+0.005$ ($[+0.002, +0.008]$, both seeds), with CG-Net boxes by $+0.008\text{--}0.010$. Trained in hybrid mode, the decoder improves SAM with MPG hybrid prompts by $+0.005\text{--}0.006$ ($[+0.004, +0.007]$), to $0.375\text{--}0.376$, which is still below the MPG mask itself (-0.003 to -0.004 , $[-0.006, -0.000]$). This is the same gain as the adaptation to out-of-fold MPG prompts alone.

- *The decoder specialises to its prompt mode.* The hybrid-trained decoder is much worse with box-only prompts (MPG boxes: -0.035 to -0.050), the box-trained one with hybrid prompts (-0.006 to -0.010).
- *Synthetic corruption alone does not help.* Without prompter prompts, the validation IoU with MPG prompts falls in every epoch, so checkpoint selection keeps the Phase-1 decoder (epoch 0). The decoder of the last epoch is worse with real prompts (-0.010 with MPG hybrid prompts, -0.076 with MPG boxes), while on ground-truth prompts of its own mode it is unchanged (box + mask: 0.881 versus 0.877 ; Table 4.8). Hand-designed corruptions do not match the errors of a real prompter.

Training the decoder for robustness therefore works as intended on the synthetic errors but does not lift SAM above its prompter: the real prompter errors that remain after training (missed cyclones and the extent of rivers, Section 4.3.4) cannot be corrected from a prompt that lacks the information.

4.3.6 Evaluation-only Improvements

Three standard evaluation-time techniques were tested on top of the trained prompters, with every free parameter chosen on the 40 validation images (Table 4.9): averaging the logits of the three seeds (*ensemble*), a per-class logit threshold instead of zero (*calibrated*), and object-level post-processing with a minimum blob size per class and a maximum latitude for TCs (*post-processed*). Finally, SAM’s own output was fed back as the next prompt for up to three rounds, SAM’s native interactive refinement loop.

- *Seed ensembles* help every non-linear prompter by 0.006 – 0.011 mean FG IoU, and end-to-end MPG, whose seeds differ most, by 0.015 (MPG: $0.379 \rightarrow 0.385$, 95% CI of the gain over seed 0 $[+0.001, +0.010]$; MSF: $0.368 \rightarrow 0.376$; CG-Net from scratch: $0.369 \rightarrow 0.380$). For the SAM-feature prompters the gain comes mostly from TCs ($+0.007$ to $+0.011$ against $+0.004$ to $+0.007$ for ARs), for CG-Net from both classes (TC $+0.009$, AR $+0.014$). The linear probe gains nothing, as its three seeds converge to nearly the same classifier.
- *Threshold calibration and post-processing* do not transfer from the validation to the test images. The thresholds selected on the 40 validation images are far from zero for TCs (logits between -1.0 and $+2.75$, depending on the prompter), but on the test set they change the IoU by only -0.005 to $+0.002$. With so few validation images the selection mostly fits noise; the unbiased threshold of the loss is the better choice.
- *Iterative SAM refinement* lowers the IoU in every round (MPG: $0.384 \rightarrow 0.369 \rightarrow 0.366 \rightarrow 0.359$). Each round loses a little AR extent and some TCs and recovers nothing, which is the prompt analysis of Section 4.3.1 applied repeatedly.
- *Ensembles across prompter families.* Averaging the logits of the three MPG and the three CG-Net (from scratch) models gives the best mask of the study, TC 0.363 / AR 0.416 (mean FG IoU 0.389 ; 0.391 with calibrated thresholds). Relative to the MPG ensemble the gain lies in the TCs ($+0.010$, CI $[-0.005, +0.024]$), relative to the CG-Net ensemble in both classes ($+0.010$ mean FG IoU, $[+0.004, +0.016]$). Prompters on the SAM features and on the raw fields thus make partly different TC errors. The comparison with the three-model ensembles is not entirely fair (six models against three), and the gain over the MPG ensemble is not significant ($+0.004$, $[-0.005, +0.012]$).

4.3.7 Prompt-free SAM with Learned Static Prompts

The most minimal prompt-free variant needs no prompter: K sparse prompt tokens per class (initialised from SAM’s positive-point embedding) are learned and fed to the frozen decoder for

Prompter	Variant	TC IoU	AR IoU	Mean FG IoU	Δ FG vs. seed 0 [95% CI]
CG-Net (trained from scratch on 358 images)	single seed (mean of seeds)	0.343	0.394	0.369	–
	single seed, calibrated (mean of seeds)	0.342	0.397	0.370	–
	ensemble	0.352	0.408	0.380	+0.005 [-0.002, +0.012]
	ensemble, calibrated	0.354	0.410	0.382	+0.007 [+0.001, +0.013]
	ensemble, calibrated, post-processed	0.351	0.410	0.381	+0.006 [-0.003, +0.015]
	SAM hybrid, round 1	0.343	0.392	0.368	-0.007 [-0.018, +0.003]
	SAM hybrid, round 2	0.334	0.394	0.364	-0.011 [-0.022, +0.000]
	SAM hybrid, round 3	0.323	0.394	0.359	-0.016 [-0.028, -0.003]
	thresholds TC +1.00 / AR -0.75; min. blob 800 / 800 px; max. TC latitude 90°				
Logistic regression, ViT block 12	single seed (mean of seeds)	0.301	0.385	0.343	–
	single seed, calibrated (mean of seeds)	0.300	0.385	0.343	–
	ensemble	0.302	0.386	0.344	-0.001 [-0.002, +0.001]
	ensemble, calibrated	0.302	0.385	0.344	-0.001 [-0.003, +0.002]
	ensemble, calibrated, post-processed	0.291	0.390	0.341	-0.004 [-0.016, +0.009]
	SAM hybrid, round 1	0.292	0.368	0.330	-0.014 [-0.027, -0.002]
	SAM hybrid, round 2	0.285	0.368	0.326	-0.018 [-0.031, -0.005]
	SAM hybrid, round 3	0.275	0.368	0.322	-0.023 [-0.037, -0.008]
	thresholds TC +0.25 / AR +0.25; min. blob 800 / 1600 px; max. TC latitude 45°				
Multi-scale fusion	single seed (mean of seeds)	0.333	0.403	0.368	–
	single seed, calibrated (mean of seeds)	0.333	0.404	0.369	–
	ensemble	0.344	0.408	0.376	+0.011 [+0.006, +0.017]
	ensemble, calibrated	0.342	0.408	0.375	+0.010 [+0.005, +0.015]
	ensemble, calibrated, post-processed	0.342	0.409	0.375	+0.011 [+0.006, +0.016]
	SAM hybrid, round 1	0.336	0.385	0.361	-0.004 [-0.010, +0.002]
	SAM hybrid, round 2	0.329	0.389	0.359	-0.006 [-0.013, +0.000]
	SAM hybrid, round 3	0.321	0.389	0.355	-0.010 [-0.018, -0.002]
	thresholds TC +2.75 / AR +0.00; min. blob 50 / 400 px; max. TC latitude 90°				
Multi-scale fusion + token gate	single seed (mean of seeds)	0.324	0.399	0.362	–
	single seed, calibrated (mean of seeds)	0.323	0.401	0.362	–
	ensemble	0.333	0.403	0.368	+0.006 [+0.003, +0.009]
	ensemble, calibrated	0.329	0.405	0.367	+0.005 [+0.001, +0.009]
	ensemble, calibrated, post-processed	0.319	0.405	0.362	+0.000 [-0.010, +0.011]
	SAM hybrid, round 1	0.316	0.391	0.354	-0.008 [-0.020, +0.003]
	SAM hybrid, round 2	0.308	0.393	0.351	-0.011 [-0.024, +0.001]
	SAM hybrid, round 3	0.298	0.392	0.345	-0.017 [-0.031, -0.004]
	thresholds TC +1.75 / AR -0.25; min. blob 800 / 400 px; max. TC latitude 90°				
Mask-prompt generator (segmentation loss)	single seed (mean of seeds)	0.344	0.413	0.379	–
	single seed, calibrated (mean of seeds)	0.343	0.412	0.378	–
	ensemble	0.353	0.418	0.385	+0.006 [+0.001, +0.010]
	ensemble, calibrated	0.349	0.417	0.383	+0.003 [-0.002, +0.009]
	ensemble, calibrated, post-processed	0.349	0.418	0.384	+0.004 [-0.004, +0.012]
	SAM hybrid, round 1	0.343	0.396	0.369	-0.010 [-0.020, +0.000]
	SAM hybrid, round 2	0.334	0.397	0.366	-0.014 [-0.025, +0.003]
	SAM hybrid, round 3	0.321	0.396	0.359	-0.021 [-0.033, -0.009]
	thresholds TC +2.00 / AR +0.25; min. blob 800 / 200 px; max. TC latitude 90°				
Mask-prompt generator (segmentation loss) + CG-Net (trained from scratch on 358 images)	single seed (mean of seeds)	0.344	0.404	0.374	–
	single seed, calibrated (mean of seeds)	0.342	0.405	0.374	–
	ensemble	0.363	0.416	0.389	+0.010 [-0.000, +0.019]
	ensemble, calibrated	0.365	0.417	0.391	+0.011 [+0.002, +0.020]
	ensemble, calibrated, post-processed	0.358	0.417	0.388	+0.008 [-0.004, +0.019]
	SAM hybrid, round 1	0.351	0.389	0.370	-0.010 [-0.023, +0.002]
	SAM hybrid, round 2	0.343	0.391	0.367	-0.013 [-0.026, +0.000]
	SAM hybrid, round 3	0.333	0.392	0.362	-0.018 [-0.031, -0.004]
	thresholds TC +0.75 / AR -0.25; min. blob 800 / 200 px; max. TC latitude 90°				
Mask-prompt generator (end-to-end via SAM)	single seed (mean of seeds)	0.335	0.378	0.356	–
	single seed, calibrated (mean of seeds)	0.331	0.383	0.357	–
	ensemble	0.342	0.400	0.371	+0.014 [+0.009, +0.019]
	ensemble, calibrated	0.338	0.400	0.369	+0.012 [+0.007, +0.017]
	ensemble, calibrated, post-processed	0.337	0.401	0.369	+0.013 [+0.005, +0.020]
	SAM hybrid, round 1	0.330	0.404	0.367	+0.010 [+0.002, +0.018]
	SAM hybrid, round 2	0.325	0.402	0.363	+0.007 [-0.002, +0.016]
	SAM hybrid, round 3	0.316	0.400	0.358	+0.001 [-0.008, +0.011]
	thresholds TC +1.50 / AR +0.00; min. blob 800 / 1600 px; max. TC latitude 90°				
Mask-prompt generator (two-stage)	single seed (mean of seeds)	0.338	0.409	0.374	–
	single seed, calibrated (mean of seeds)	0.338	0.412	0.375	–
	ensemble	0.349	0.415	0.382	+0.009 [+0.003, +0.015]
	ensemble, calibrated	0.347	0.417	0.382	+0.009 [+0.003, +0.016]
	ensemble, calibrated, post-processed	0.343	0.418	0.380	+0.008 [+0.000, +0.015]
	SAM hybrid, round 1	0.337	0.395	0.366	-0.007 [-0.017, +0.005]
	SAM hybrid, round 2	0.329	0.398	0.364	-0.009 [-0.020, +0.002]
	SAM hybrid, round 3	0.319	0.398	0.358	-0.014 [-0.026, -0.002]
	thresholds TC +1.75 / AR +1.25; min. blob 800 / 200 px; max. TC latitude 90°				
Mask-prompt generator (segmentation loss, label smoothing)	single seed (mean of seeds)	0.338	0.412	0.375	–
	single seed, calibrated (mean of seeds)	0.338	0.412	0.375	–
	ensemble	0.349	0.418	0.383	+0.013 [+0.009, +0.018]
	ensemble, calibrated	0.348	0.417	0.383	+0.013 [+0.008, +0.017]
	ensemble, calibrated, post-processed	0.351	0.418	0.384	+0.015 [+0.009, +0.020]
	SAM hybrid, round 1	0.341	0.398	0.370	-0.000 [-0.008, +0.008]
	SAM hybrid, round 2	0.332	0.398	0.365	-0.005 [-0.014, +0.004]
	SAM hybrid, round 3	0.320	0.397	0.359	-0.011 [-0.021, -0.001]
	thresholds TC +0.75 / AR +0.25; min. blob 400 / 1600 px; max. TC latitude 90°				
Mask-prompt generator + raw CG-Net fields	single seed (mean of seeds)	0.340	0.409	0.375	–
	single seed, calibrated (mean of seeds)	0.339	0.411	0.375	–
	ensemble	0.347	0.416	0.382	+0.011 [+0.007, +0.016]
	ensemble, calibrated	0.344	0.416	0.380	+0.010 [+0.004, +0.015]
	ensemble, calibrated, post-processed	0.347	0.418	0.382	+0.012 [+0.004, +0.020]
	SAM hybrid, round 1	0.340	0.397	0.369	-0.001 [-0.011, +0.009]
	SAM hybrid, round 2	0.333	0.398	0.365	-0.005 [-0.016, +0.006]
	SAM hybrid, round 3	0.321	0.396	0.358	-0.012 [-0.024, +0.001]
	thresholds TC +1.75 / AR +0.25; min. blob 800 / 1600 px; max. TC latitude 90°				

Table 4.9: Evaluation-only improvements of the prompter masks (test set). Ensemble: mean of the logits of the three seeds; calibrated: per-class logit threshold chosen on the validation images; post-processed: minimum blob size per class and maximum TC latitude chosen on the validation images; SAM hybrid round k : the post-processed mask as hybrid prompt, then SAM’s own output fed back as the prompt. Δ : paired bootstrap against the seed-0 prompter mask.

every image, so all image-specific information must come from the decoder’s attention to the image embedding. With $K = 4$ this reaches TC 0.237 ± 0.001 / AR 0.324 ± 0.008 (three seeds), with $K = 16$ TC 0.248 / AR 0.329 , i.e. 0.09–0.10 mean FG IoU below MPG. Object recall is high (TC 0.87, AR 0.98), but the output breaks into thousands of fragments (object precision about 0.2): a fixed query cannot decide *which* humid, windy region is a TC or an AR. This decision needs an image-dependent prompter.

4.3.8 Robustness to the Frozen Checkpoint

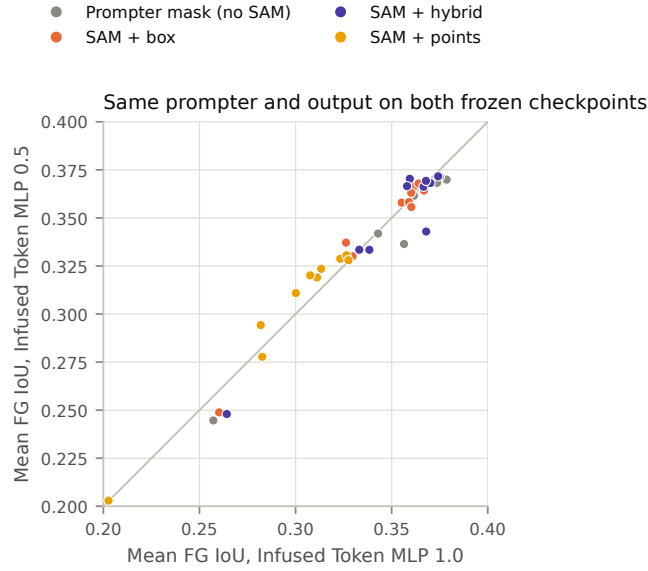


Figure 4.14: Mean FG IoU of every prompter and output on the MLP-1.0 (x) and MLP-0.5 (y) checkpoints. Diagonal: equal performance.

All experiments were repeated with the MLP-0.5 checkpoint (Table A.5; MSF models with one seed). The results lie close to the diagonal (Figure 4.14), so the conclusions do not depend on the checkpoint. On this checkpoint MPG (0.370) ties MSF (0.370) and the smoothed MPG (0.371); all three stay ahead of CG-Net (0.366; AR +0.027 to +0.028) and of the linear probe (0.342), and again no SAM output exceeds the best prompter mask (best SAM output 0.373 versus 0.371, within noise).

4.3.9 Qualitative Examples

Figure 4.15 illustrates the quantitative findings. All prompters find the long rivers of the southern hemisphere and the large cyclone over the Gulf of Mexico; they differ in the extent of the rivers and in small false or missed objects. SAM with boxes (e, g) fills every box with a river-shaped mask, including the false AR over the central North Pacific that the prompters hallucinated, and cuts rivers at the box borders; the hybrid prompt (f) follows the prompter’s shape closely. The learned static prompts (h) mark every moist band. A second example is shown in Figure A.1 (Appendix).

4.3.10 Summary

- **RQ1.** With perfect prompts the Phase-1 SAM reaches TC 0.87 / AR 0.92 (hybrid prompts). It does not correct prompts: missed objects stay missed, false prompts become false objects, and 10–20% box error costs 0.1–0.3 IoU.

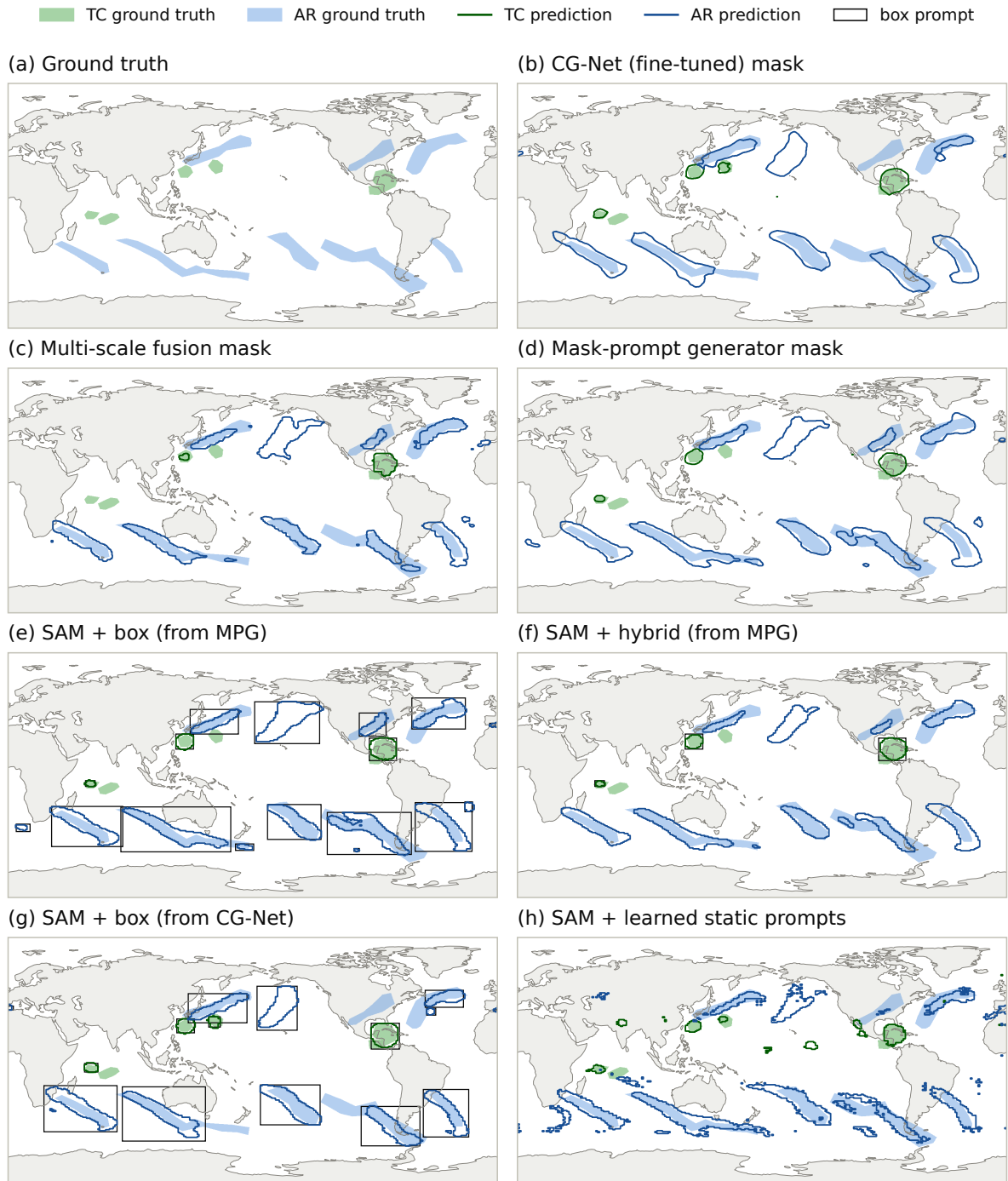


Figure 4.15: Test image 17. Filled: ground truth (TC green, AR blue); outlines: prediction; black rectangles: box prompts. (b–d) prompter masks; (e, f) SAM prompted by MPG with boxes and with hybrid prompts; (g) SAM prompted by CG-Net boxes; (h) SAM with learned static prompts.

- **RQ2.** The SAM rows of the earlier Table 4.13 do not reproduce; SAM prompted by CG-Net is slightly worse than CG-Net for every prompt type. The ranking of the prompt types is confirmed.
- **RQ3.** The mask-prompt generator (0.67 M parameters, 5.7 GFLOPs) is the best prompter: TC 0.344 / AR 0.413, AR +0.031 over the stored fine-tuned CG-Net and +0.011 mean FG IoU over the multi-scale fusion generator at 122× fewer FLOPs. Against CG-Net re-trained under the same protocol it is better only for ARs (+0.013 to +0.018); the overall difference is not significant. An ensemble of both families gives the best mask (0.389).
- **RQ4.** SAM prompted by any automatic prompter is at best as good as that prompter. Neither training the prompter through SAM nor adapting the decoder to realistic prompts changes this.
- **RQ5.** ARs are found but their extent is wrong; TCs are missed and falsely detected. A prompt-conditioned decoder cannot repair either error.

Chapter 5

Discussion

5.1 Discussion of Automatic Prompting

5.1.1 Why SAM Cannot Improve on Its Prompter

The central finding of Section 4.3 is that SAM with automatic prompts is bounded by its prompter. This follows from what the Phase-1 decoder was trained to do. It learned $p(\text{mask} \mid \text{image}, \text{prompt})$ from prompts that were always correct: every box contained exactly one object, and the extent of the box was the extent of the object. Under this training distribution the optimal decoder *trusts* its prompt: it segments whatever lies inside a box and never has a reason to output nothing or to extend a mask beyond its prompt. The prompt analysis confirms this behaviour directly (Figure 4.3). The errors of a prompter are of exactly these kinds, namely false objects, missed objects and wrong AR extents (Figures 4.10 and 4.11), so they pass through SAM unchanged. What SAM can add is boundary quality inside a correct prompt; but for ARs the dominant error is the extent of the whole river, not its boundary pixels, and for TCs the objects are small enough that detection dominates.

The decoder-adaptation experiments show that this is not only a question of the training distribution. Even a decoder trained on realistic, out-of-fold prompts, or on deliberately corrupted prompts with empty targets for false boxes, becomes more tolerant of box errors and rejects some false prompts, but it cannot add information that the prompt does not contain. Prompter and decoder read the same frozen image embedding; whatever the decoder could infer about the extent of an AR from that embedding, a prompter trained directly on the labels infers as well.

A further reason lies in the labels. ClimateNet’s labels are the consensus of several experts, and the mean performance of a single expert against this consensus is AR 0.341 / TC 0.257 IoU (Table 2.1). All good prompters of this study (TC 0.33–0.35, AR 0.38–0.41) already exceed the average single expert. The remaining errors, such as the extent of a river, where it starts and ends, or weak cyclones, are precisely where experts disagree, so part of the gap to the oracle is irreducible label ambiguity rather than a deficiency of the model.

5.1.2 Why a Small Generator Beats the Larger Ones

The encoder describes the image on a 64×64 grid of 16×16 -pixel patches. The ground truth has a resolution of 768×1152 , but its information content at object boundaries is limited by the ambiguity between experts. MSF upsamples the features to 1024×1024 by nearest-neighbour interpolation and runs most of its 702 GFLOPs there, which cannot create information and makes the model slow to train. MPG spends its capacity where the information is, at 64×64 , with a receptive field of about 15 patches from the dilated convolutions on top of the encoder’s own global attention, and uses two learned upsampling steps only to reach the 256×256 resolution of SAM’s dense prompt. The strong linear probe on the last ViT block supports this view: the

adapted encoder has already done most of the work, and the prompter mainly aggregates context and removes noise.

5.1.3 Lessons for Prompt Design

1. If sparse prompts are used, they should be tight boxes derived from a segmentation. Points, loose boxes and boxes regressed by a detector are harmful.
2. Dense prompts are more robust to prompt errors and carry the shape of an AR, but they must be passed as logits in the value range the decoder was trained with, and small TCs need an accompanying box on checkpoints whose decoder ignores weak dense evidence.
3. One dense prompt per class is sufficient. This makes SAM decoding cheap: one decoder call per class and image instead of one per object.

5.1.4 Implications for a Prompt-free ClimateSAM

For a prompt-free system, the prompter’s own mask is the natural output (MPG: TC 0.344 / AR 0.413, mean IoU 0.567 including background). SAM with hybrid prompts gives masks of statistically similar quality (-0.008 mean FG IoU) with smoother boundaries, at the cost of two decoder calls per image. The contribution of the foundation model is thus its *encoder*: the adapted features allow a 0.67M-parameter head to match a dedicated segmentation network trained under the same protocol on the raw fields (mean FG IoU 0.379 versus 0.369–0.375, better for ARs by 0.013–0.018), and a linear probe to come within about 0.03 of it. In terms of cost, however, CG-Net needs 23 GFLOPs per image, the frozen encoder plus MPG about 980 GFLOPs. For a purely automatic system, a well-trained small network is therefore the cheaper choice at nearly the same quality, and the two are complementary: their ensemble is the best automatic mask of this study (0.389). The promptable decoder remains valuable for interactive use, where a human annotator provides the boxes (Table 4.12: TC 0.72 / AR 0.63–0.64 with ground-truth boxes), but not as a refinement stage after an automatic prompter.

5.1.5 Limitations

- *Small test set.* With 61 test images, differences below about 0.01 IoU lie within the bootstrap intervals and should not be interpreted. Checkpoints are selected on only 40 validation images drawn from the same years as the training images.
- *Seeds.* Three seeds per configuration on the primary checkpoint; on the second checkpoint the MSF models were trained with one seed. The seed variance (standard deviation 0.001–0.01) is small compared with the variance from sampling test images.
- *Two frozen checkpoints.* Both are Infused-Token ViT-B models; LoRA-adapted and ViT-L encoders were not tested in Phase 2.
- *External prompter.* The fine-tuned CG-Net was trained on all 398 training images, including the validation images of this protocol; this affects only experiments that use its outputs on training images. The CG-Net re-trained from the official ClimateNet weights inherits this for its checkpoint selection (not for its test numbers); the CG-Net trained from scratch is unaffected.
- *Object metrics.* An object counts as found if one pixel overlaps; stricter matching ($\text{IoU} \geq 0.5$) would lower all recalls and precisions but not change the comparison.
- *Earlier Table 4.13.* The re-run shows that its SAM rows do not reproduce with the current code and checkpoints; their origin could not be identified.

5.1.6 Future Work

1. *Ensembles and fusion of prompter families.* The ensemble of MPG and CG-Net is the best automatic mask; a larger same-family ensemble is the missing control, and a fusion that keeps the diversity of the two input types (unlike MPG with raw fields) may keep the gain at lower cost.
2. *Realistic prompts already in Phase 1.* The prompt-robust decoder was trained only in Phase 2 with a frozen prompt encoder. Training Phase 1 with corrupted and prompter prompts is the stronger version; given Section 4.3.5 its benefit for automatic prompting is likely small, but it would give a more forgiving interactive tool.
3. *Query-based prompt-free decoding.* Replacing static prompts by image-dependent object queries (DETR/Mask2Former style) with a matching-based loss turns SAM into an instance segmenter and addresses the TC detection errors directly.
4. *Joint fine-tuning of encoder adapters and prompter.* Unfreezing the infused-token adapters together with MPG on the segmentation loss lets the encoder adapt to the prompt-free objective.
5. *Longitudinal periodicity.* Circular padding in the prompter and test-time averaging over longitudinally rolled inputs would fix objects cut at the date line.
6. *Stricter object metrics.* Object recall and precision at $\text{IoU} \geq 0.3 / 0.5$ and event counts per image would make the detection analysis comparable with the ClimateNet literature.

Appendix A

Automatic Prompting: Additional Material

A.1 Complete Results of the Automatic-Prompting Benchmark

Prompter	Prompter mask (no SAM)	SAM + box	SAM + box (+10%)	SAM + points	SAM + box + points	SAM + mask logits	SAM + box + mask	SAM + hybrid	mean(prompter, SAM hybrid)	SAM + learned static prompts
Ground truth (oracle)	1.00 / 1.00	0.72 / 0.62	0.59 / 0.57	0.53 / 0.58	0.58 / 0.60	0.77 / 0.92	0.87 / 0.88	0.87 / 0.92	0.95 / 0.98	–
CG-Net (official weights)	0.33 / 0.33	0.32 / 0.34	0.28 / 0.34	0.25 / 0.31	0.28 / 0.32	0.33 / 0.34	0.33 / 0.34	0.33 / 0.34	–	–
CG-Net (fine-tuned)	0.35 / 0.38	0.34 / 0.37	0.31 / 0.36	0.28 / 0.34	0.32 / 0.36	0.34 / 0.38	0.35 / 0.38	0.35 / 0.38	–	–
CG-Net (re-trained on 358 images, official init.)	0.35 / 0.40	0.35 / 0.39	0.33 / 0.38	0.29 / 0.36	0.32 / 0.37	0.34 / 0.40	0.35 / 0.40	0.35 / 0.40	–	–
CG-Net (re-trained from scratch on 358 images)	0.34 / 0.39	0.34 / 0.38	0.33 / 0.37	0.31 / 0.35	0.33 / 0.37	0.32 / 0.40	0.34 / 0.39	0.34 / 0.40	–	–
Logistic regression, ViT block 1	0.20 / 0.32	0.20 / 0.32	0.19 / 0.32	0.14 / 0.27	0.18 / 0.30	0.04 / 0.32	0.20 / 0.31	0.21 / 0.32	0.21 / 0.32	–
Logistic regression, ViT block 12	0.30 / 0.39	0.29 / 0.36	0.28 / 0.35	0.25 / 0.31	0.28 / 0.35	0.04 / 0.37	0.29 / 0.36	0.30 / 0.37	0.30 / 0.37	–
Multi-scale fusion	0.33 / 0.40	0.34 / 0.38	0.33 / 0.38	0.30 / 0.35	0.33 / 0.37	0.31 / 0.39	0.33 / 0.38	0.33 / 0.39	0.33 / 0.39	–
Multi-scale fusion + token gate	0.32 / 0.40	0.34 / 0.38	0.32 / 0.38	0.28 / 0.34	0.32 / 0.37	0.26 / 0.39	0.33 / 0.38	0.33 / 0.39	0.33 / 0.39	–
Multi-scale fusion + token gate, CG blocks	0.32 / 0.35	0.29 / 0.34	0.27 / 0.30	0.26 / 0.28	0.28 / 0.30	0.18 / 0.39	0.31 / 0.38	0.32 / 0.39	0.32 / 0.39	–
Multi-scale fusion + token gate, shared weights	0.33 / 0.40	0.34 / 0.39	0.32 / 0.38	0.29 / 0.34	0.33 / 0.37	0.23 / 0.39	0.33 / 0.39	0.33 / 0.39	0.33 / 0.39	–
Mask-prompt generator (segmentation loss)	0.34 / 0.41	0.34 / 0.39	0.32 / 0.38	0.30 / 0.35	0.32 / 0.37	0.30 / 0.40	0.34 / 0.39	0.34 / 0.40	0.34 / 0.40	–
Mask-prompt generator (end-to-end via SAM)	0.33 / 0.38	0.34 / 0.38	0.32 / 0.38	0.29 / 0.31	0.32 / 0.36	0.30 / 0.40	0.33 / 0.38	0.33 / 0.40	0.33 / 0.40	–
Mask-prompt generator (two-stage)	0.34 / 0.41	0.33 / 0.39	0.31 / 0.36	0.29 / 0.33	0.32 / 0.36	0.29 / 0.41	0.33 / 0.40	0.34 / 0.41	0.34 / 0.41	–
Mask-prompt generator (segmentation loss, label smoothing)	0.34 / 0.41	0.34 / 0.39	0.32 / 0.37	0.31 / 0.35	0.33 / 0.37	0.29 / 0.40	0.33 / 0.40	0.33 / 0.40	0.34 / 0.40	–
Mask-prompt generator + raw CG-Net fields	0.34 / 0.41	0.34 / 0.39	0.32 / 0.37	0.30 / 0.35	0.32 / 0.37	0.30 / 0.40	0.34 / 0.39	0.34 / 0.40	0.34 / 0.40	–
Box head on SAM features (YOLO-style)	0.09 / 0.16	0.11 / 0.30	–	–	–	–	–	–	–	–
Learned static prompts (4 tokens / class)	–	–	–	–	–	–	–	–	–	0.24 / 0.32
Learned static prompts (16 tokens / class)	–	–	–	–	–	–	–	–	–	0.25 / 0.33

Table A.1: TC IoU / AR IoU for every way of turning the prompter output into SAM prompts (Infused Token, MLP ratio 1.0).

A.2 Issues Found in the Original Phase-2 Code

1. `test_prompt_effect.py` and the Phase-1 validation in `train_adaptation.py` encode images with `ClimateSAM.set_infer_img()`, which feeds the first three raw channels to the encoder and skips the learned input adapter (`encode_images()` applies it). Effect: -0.00 to -0.03 IoU for the linear adapter, -0.04 to -0.13 mean FG IoU for the nonlinear adapter (Section A.3).
2. The multi-scale generators predict three channels, of which only TC and AR are supervised (as independent sigmoids) by `compute_generator_loss`; the untrained background channel takes part in the final argmax.
3. `LogisticRegressionPrompter` classifies the output of the first ViT block instead of the last.
4. `StreamSegMetrics.update(gt, pred)` is called with swapped arguments in several scripts. IoU is symmetric and unaffected; mean and frequency-weighted accuracies are affected.
5. All original Phase-2 training scripts select the best epoch on the test set.
6. The official ClimateNet CG-Net checkpoint predicts background everywhere in evaluation mode, because its stored BatchNorm running statistics do not match its weights; it works with batch statistics.

Comparison (B minus A)	Δ TC IoU [95% CI]	Δ AR IoU [95% CI]	Δ mean FG IoU [95% CI]	P(Δ FG>0)
CG-Net (official weights): SAM + box vs. prompter mask	-0.009 [-0.017, -0.002]	+0.009 [+0.004, +0.014]	-0.000 [-0.005, +0.004]	0.45
CG-Net (official weights): SAM + hybrid vs. prompter mask	+0.000 [-0.003, +0.003]	+0.006 [+0.005, +0.007]	+0.003 [+0.002, +0.005]	1.00
CG-Net (fine-tuned): SAM + box vs. prompter mask	-0.008 [-0.017, +0.001]	-0.013 [-0.018, -0.006]	-0.010 [-0.016, -0.004]	0.00
CG-Net (fine-tuned): SAM + hybrid vs. prompter mask	-0.001 [-0.004, +0.003]	+0.003 [+0.002, +0.004]	+0.001 [-0.001, +0.003]	0.89
CG-Net (re-trained on 358 images, official init.): SAM + box vs. prompter mask	-0.005 [-0.014, +0.004]	-0.013 [-0.019, -0.007]	-0.009 [-0.014, -0.004]	0.00
CG-Net (re-trained on 358 images, official init.): SAM + hybrid vs. prompter mask	-0.002 [-0.005, +0.000]	+0.001 [-0.000, +0.001]	-0.001 [-0.002, +0.000]	0.99
CG-Net (trained from scratch on 358 images): SAM + box vs. prompter mask	+0.000 [-0.008, +0.007]	-0.015 [-0.021, -0.009]	-0.007 [-0.012, -0.002]	0.00
CG-Net (trained from scratch on 358 images): SAM + hybrid vs. prompter mask	-0.004 [-0.006, -0.001]	+0.001 [+0.000, +0.002]	-0.001 [-0.002, -0.000]	0.02
Logistic regression, ViT block 1: SAM + box vs. prompter mask	+0.005 [-0.003, +0.012]	+0.001 [-0.003, +0.006]	+0.003 [-0.002, +0.007]	0.90
Logistic regression, ViT block 1: SAM + hybrid vs. prompter mask	+0.017 [+0.008, +0.026]	-0.003 [-0.014, +0.007]	+0.007 [-0.000, +0.013]	0.97
Logistic regression, ViT block 1: mean(prompter, SAM hybrid) vs. prompter mask	+0.017 [+0.009, +0.025]	-0.003 [-0.013, +0.007]	+0.007 [+0.001, +0.013]	0.98
Logistic regression, ViT block 12: SAM + box vs. prompter mask	-0.010 [-0.029, +0.009]	-0.024 [-0.029, -0.019]	-0.017 [-0.027, -0.006]	0.00
Logistic regression, ViT block 12: SAM + hybrid vs. prompter mask	+0.003 [-0.005, +0.011]	-0.012 [-0.024, -0.001]	-0.005 [-0.012, -0.003]	0.11
Logistic regression, ViT block 12: mean(prompter, SAM hybrid) vs. prompter mask	+0.004 [-0.004, +0.011]	-0.011 [-0.023, +0.000]	-0.004 [-0.011, +0.003]	0.15
Multi-scale fusion: SAM + box vs. prompter mask	+0.006 [-0.004, +0.015]	-0.018 [-0.023, -0.013]	-0.006 [-0.012, -0.000]	0.01
Multi-scale fusion: SAM + hybrid vs. prompter mask	-0.003 [-0.007, +0.001]	-0.014 [-0.023, -0.005]	-0.009 [-0.013, -0.004]	0.00
Multi-scale fusion: mean(prompter, SAM hybrid) vs. prompter mask	+0.000 [-0.003, +0.003]	-0.012 [-0.020, -0.004]	-0.006 [-0.010, -0.002]	0.00
Multi-scale fusion + token gate: SAM + box vs. prompter mask	+0.011 [+0.000, +0.022]	-0.017 [-0.022, -0.011]	-0.003 [-0.009, +0.004]	0.22
Multi-scale fusion + token gate: SAM + hybrid vs. prompter mask	+0.005 [-0.000, +0.010]	-0.012 [-0.021, -0.003]	-0.004 [-0.009, +0.002]	0.09
Multi-scale fusion + token gate: mean(prompter, SAM hybrid) vs. prompter mask	+0.005 [+0.002, +0.009]	-0.010 [-0.018, -0.002]	-0.002 [-0.007, +0.002]	0.14
Multi-scale fusion + token gate, CG blocks: SAM + box vs. prompter mask	-0.036 [-0.058, -0.013]	-0.019 [-0.024, -0.015]	-0.027 [-0.040, -0.016]	0.00
Multi-scale fusion + token gate, CG blocks: SAM + hybrid vs. prompter mask	-0.003 [-0.008, +0.003]	+0.037 [+0.031, +0.044]	+0.017 [+0.013, +0.022]	1.00
Multi-scale fusion + token gate, CG blocks: mean(prompter, SAM hybrid) vs. prompter mask	-0.001 [-0.006, +0.004]	+0.037 [+0.031, +0.043]	+0.018 [+0.014, +0.022]	1.00
Multi-scale fusion + token gate, shared weights: SAM + box vs. prompter mask	+0.006 [-0.004, +0.016]	-0.017 [-0.022, -0.012]	-0.005 [-0.012, +0.001]	0.04
Multi-scale fusion + token gate, shared weights: SAM + hybrid vs. prompter mask	+0.001 [-0.006, +0.007]	-0.011 [-0.021, -0.002]	-0.005 [-0.011, +0.000]	0.03
Multi-scale fusion + token gate, shared weights: mean(prompter, SAM hybrid) vs. prompter mask	+0.002 [-0.003, +0.007]	-0.009 [-0.018, -0.000]	-0.003 [-0.009, +0.001]	0.09
Mask-prompt generator (segmentation loss): SAM + box vs. prompter mask	-0.002 [-0.009, +0.006]	-0.022 [-0.028, -0.016]	-0.012 [-0.017, -0.007]	0.00
Mask-prompt generator (segmentation loss): SAM + hybrid vs. prompter mask	-0.002 [-0.007, +0.003]	-0.015 [-0.024, -0.006]	-0.008 [-0.014, -0.003]	0.00
Mask-prompt generator (segmentation loss): mean(prompter, SAM hybrid) vs. prompter mask	-0.000 [-0.005, +0.005]	-0.013 [-0.021, -0.004]	-0.006 [-0.011, -0.001]	0.01
Mask-prompt generator (end-to-end via SAM): SAM + box vs. prompter mask	+0.001 [-0.008, +0.009]	+0.006 [+0.001, +0.012]	+0.004 [-0.002, +0.009]	0.91
Mask-prompt generator (end-to-end via SAM): SAM + hybrid vs. prompter mask	-0.001 [-0.005, +0.003]	+0.024 [+0.021, +0.028]	+0.012 [+0.009, +0.015]	1.00
Mask-prompt generator (end-to-end via SAM): mean(prompter, SAM hybrid) vs. prompter mask	-0.000 [-0.004, +0.003]	+0.024 [+0.020, +0.028]	+0.012 [+0.009, +0.015]	1.00
Mask-prompt generator (two-stage): SAM + box vs. prompter mask	-0.004 [-0.013, +0.004]	-0.023 [-0.028, -0.017]	-0.014 [-0.019, -0.008]	0.00
Mask-prompt generator (two-stage): SAM + hybrid vs. prompter mask	-0.001 [-0.007, +0.005]	+0.002 [-0.006, +0.010]	+0.001 [-0.005, +0.006]	0.58
Mask-prompt generator (two-stage): mean(prompter, SAM hybrid) vs. prompter mask	+0.001 [-0.004, +0.006]	+0.003 [-0.004, +0.011]	+0.002 [-0.003, +0.007]	0.80
Mask-prompt generator (segmentation loss, label smoothing): SAM + box vs. prompter mask	-0.001 [-0.006, +0.009]	-0.024 [-0.029, -0.018]	-0.011 [-0.016, -0.006]	0.00
Mask-prompt generator (segmentation loss, label smoothing): SAM + hybrid vs. prompter mask	-0.003 [-0.009, +0.002]	-0.011 [-0.019, -0.002]	-0.007 [-0.013, -0.001]	0.01
Mask-prompt generator (segmentation loss, label smoothing): mean(prompter, SAM hybrid) vs. prompter mask	-0.001 [-0.005, +0.004]	-0.009 [-0.017, -0.001]	-0.005 [-0.010, -0.000]	0.02
Mask-prompt generator + raw CG-Net fields: SAM + box vs. prompter mask	+0.000 [-0.008, +0.008]	-0.023 [-0.028, -0.017]	-0.011 [-0.016, -0.007]	0.00
Mask-prompt generator + raw CG-Net fields: SAM + hybrid vs. prompter mask	-0.003 [-0.007, +0.001]	-0.010 [-0.020, -0.001]	-0.007 [-0.012, -0.002]	0.00
Mask-prompt generator + raw CG-Net fields: mean(prompter, SAM hybrid) vs. prompter mask	-0.001 [-0.004, +0.002]	-0.009 [-0.017, -0.000]	-0.005 [-0.009, -0.000]	0.01
Box head on SAM features (YOLO-style): SAM + box vs. prompter mask	+0.012 [+0.007, +0.017]	+0.147 [+0.135, +0.158]	+0.079 [+0.072, +0.086]	1.00
Mask-prompt generator (segmentation loss) vs. CG-Net (fine-tuned) (prompter masks)	-0.005 [-0.030, +0.020]	+0.031 [+0.020, +0.043]	+0.013 [-0.000, +0.027]	0.97
Mask-prompt generator (segmentation loss) vs. CG-Net (trained from scratch on 358 images) (prompter masks)	+0.001 [-0.018, +0.023]	+0.018 [+0.009, +0.027]	+0.010 [-0.002, +0.022]	0.95
Mask-prompt generator (segmentation loss) vs. CG-Net (re-trained on 358 images, official init.) (prompter masks)	-0.006 [-0.027, +0.015]	+0.013 [+0.003, +0.024]	+0.004 [-0.008, +0.016]	0.72
CG-Net (trained from scratch on 358 images) vs. CG-Net (fine-tuned) (prompter masks)	-0.007 [-0.029, +0.016]	+0.013 [+0.003, +0.023]	+0.003 [-0.009, +0.016]	0.69
Mask-prompt generator (segmentation loss) vs. Multi-scale fusion (prompter masks)	+0.011 [-0.003, +0.025]	+0.010 [+0.005, +0.016]	+0.011 [+0.003, +0.019]	0.99
Mask-prompt generator (segmentation loss) vs. Logistic regression, ViT block 12 (prompter masks)	+0.044 [+0.025, +0.062]	+0.028 [+0.020, +0.036]	+0.036 [+0.025, +0.046]	1.00
Multi-scale fusion + token gate, CG blocks vs. Multi-scale fusion + token gate (prompter masks)	-0.000 [-0.013, +0.014]	-0.046 [-0.060, -0.032]	-0.023 [-0.032, -0.015]	0.00
Multi-scale fusion + token gate, shared weights vs. Multi-scale fusion + token gate (prompter masks)	+0.008 [+0.002, +0.013]	+0.003 [+0.000, +0.006]	+0.006 [+0.003, +0.009]	1.00
Mask-prompt generator (segmentation loss) vs. Multi-scale fusion + token gate, shared weights (prompter masks)	+0.012 [-0.001, +0.026]	+0.011 [+0.005, +0.016]	+0.011 [+0.004, +0.020]	1.00
Mask-prompt generator + raw CG-Net fields vs. Mask-prompt generator (segmentation loss) (prompter masks)	-0.004 [-0.012, +0.002]	-0.004 [-0.007, -0.001]	-0.004 [-0.008, -0.001]	0.01
Mask-prompt generator + raw CG-Net fields vs. CG-Net (trained from scratch on 358 images) (prompter masks)	-0.003 [-0.023, +0.018]	+0.014 [+0.004, +0.024]	+0.006 [-0.006, +0.019]	0.82
Multi-scale fusion + token gate vs. Multi-scale fusion (prompter masks)	-0.009 [-0.015, -0.003]	-0.004 [-0.005, -0.002]	-0.006 [-0.009, -0.003]	0.00
Logistic regression, ViT block 12 vs. Logistic regression, ViT block 1 (prompter masks)	+0.105 [+0.078, +0.133]	+0.066 [+0.053, +0.080]	+0.086 [+0.070, +0.101]	1.00
Mask-prompt generator (segmentation loss, label smoothing) vs. Mask-prompt generator (segmentation loss) (prompter masks)	-0.006 [-0.012, +0.000]	-0.001 [-0.004, +0.001]	-0.004 [-0.007, -0.000]	0.01
Multi-scale fusion vs. CG-Net (fine-tuned) (prompter masks)	-0.016 [-0.041, +0.008]	+0.021 [+0.009, +0.033]	+0.003 [-0.011, +0.016]	0.64
Mask-prompt generator (two-stage): SAM + hybrid vs. segmentation-loss generator mask	-0.007 [-0.013, -0.000]	-0.002 [-0.007, +0.003]	-0.004 [-0.008, -0.000]	0.02
Mask-prompt generator (end-to-end via SAM): SAM + hybrid vs. segmentation-loss generator mask	-0.011 [-0.020, -0.001]	-0.010 [-0.018, -0.003]	-0.011 [-0.016, -0.005]	0.00
Learned static prompts (4 tokens / class) vs. segmentation-loss generator mask	-0.107 [-0.142, -0.073]	-0.089 [-0.102, -0.077]	-0.098 [-0.117, -0.079]	0.00
Learned static prompts (16 tokens / class) vs. segmentation-loss generator mask	-0.097 [-0.131, -0.063]	-0.084 [-0.095, -0.072]	-0.090 [-0.110, -0.071]	0.00

Table A.2: Paired bootstrap over the 61 test images (2000 resamples; learned prompters: the three seeds pooled). Infused Token, MLP ratio 1.0.

Prompter	Class	As is	No false objects	Add missed objects	Perfect detection	Perfect shape
CG-Net (official weights)	TC	0.327	0.453	0.405	0.562	0.571
CG-Net (fine-tuned)	TC	0.349	0.439	0.457	0.575	0.620
CG-Net (re-trained on 358 images, official init.)	TC	0.350±0.005	0.441±0.018	0.460±0.008	0.580±0.004	0.631±0.012
CG-Net (trained from scratch on 358 images)	TC	0.343±0.007	0.410±0.013	0.506±0.006	0.605±0.008	0.628±0.007
Logistic regression, ViT block 1	TC	0.228±0.000	0.324±0.008	0.304±0.005	0.432±0.004	0.593±0.015
Logistic regression, ViT block 12	TC	0.306±0.003	0.393±0.012	0.401±0.007	0.514±0.002	0.645±0.012
Multi-scale fusion	TC	0.333±0.002	0.405±0.011	0.461±0.018	0.560±0.011	0.651±0.005
Multi-scale fusion + token gate	TC	0.325±0.001	0.395±0.004	0.446±0.009	0.542±0.008	0.657±0.005
Multi-scale fusion + token gate, CG blocks	TC	0.328±0.017	0.405±0.004	0.436±0.030	0.539±0.004	0.657±0.035
Multi-scale fusion + token gate, shared weights	TC	0.332±0.006	0.405±0.015	0.450±0.005	0.548±0.008	0.656±0.012
Mask-prompt generator (segmentation loss)	TC	0.345±0.005	0.420±0.002	0.481±0.015	0.585±0.009	0.636±0.012
Mask-prompt generator (end-to-end via SAM)	TC	0.337±0.008	0.410±0.005	0.474±0.012	0.577±0.010	0.637±0.019
Mask-prompt generator (two-stage)	TC	0.340±0.005	0.415±0.013	0.470±0.003	0.573±0.014	0.641±0.019
Mask-prompt generator (segmentation loss, label smoothing)	TC	0.340±0.010	0.407±0.011	0.483±0.016	0.579±0.015	0.645±0.002
Mask-prompt generator + raw CG-Net fields	TC	0.339±0.001	0.422±0.005	0.465±0.009	0.580±0.007	0.627±0.005
CG-Net (official weights)	AR	0.333	0.396	0.348	0.414	0.761
CG-Net (fine-tuned)	AR	0.382	0.437	0.421	0.482	0.787
CG-Net (re-trained on 358 images, official init.)	AR	0.399±0.005	0.456±0.008	0.437±0.007	0.499±0.005	0.805±0.006
CG-Net (trained from scratch on 358 images)	AR	0.394±0.008	0.453±0.023	0.427±0.006	0.490±0.008	0.806±0.022
Logistic regression, ViT block 1	AR	0.319±0.001	0.396±0.004	0.324±0.003	0.402±0.002	0.744±0.021
Logistic regression, ViT block 12	AR	0.385±0.001	0.463±0.002	0.390±0.002	0.468±0.002	0.785±0.004
Multi-scale fusion	AR	0.403±0.004	0.468±0.009	0.411±0.005	0.478±0.011	0.818±0.017
Multi-scale fusion + token gate	AR	0.399±0.003	0.462±0.009	0.407±0.001	0.471±0.007	0.824±0.016
Multi-scale fusion + token gate, CG blocks	AR	0.355±0.022	0.438±0.023	0.356±0.023	0.440±0.024	0.706±0.035
Multi-scale fusion + token gate, shared weights	AR	0.402±0.001	0.469±0.004	0.411±0.002	0.479±0.002	0.815±0.009
Mask-prompt generator (segmentation loss)	AR	0.413±0.001	0.479±0.002	0.423±0.001	0.490±0.002	0.817±0.004
Mask-prompt generator (end-to-end via SAM)	AR	0.378±0.019	0.451±0.018	0.383±0.020	0.456±0.017	0.801±0.053
Mask-prompt generator (two-stage)	AR	0.409±0.002	0.485±0.003	0.415±0.003	0.491±0.002	0.789±0.013
Mask-prompt generator (segmentation loss, label smoothing)	AR	0.412±0.001	0.483±0.006	0.421±0.003	0.493±0.003	0.806±0.008
Mask-prompt generator + raw CG-Net fields	AR	0.409±0.006	0.478±0.003	0.419±0.010	0.489±0.002	0.808±0.025

Table A.3: IoU of the prompter mask after removing one error type: deleting predicted objects that touch no ground truth, adding the ground-truth objects that were missed, both, or giving every detected object its exact ground-truth shape.

Prompter	Trainable parameters	Training time (min)	Selected epoch
CG-Net (re-trained on 358 images, official init.)	494,232	27.3	7
CG-Net (trained from scratch on 358 images)	494,232	27.9	17
Logistic regression, ViT block 1	1,538	1.4	46
Logistic regression, ViT block 12	1,538	2.0	39
Multi-scale fusion	1,406,474	82.6	38
Multi-scale fusion + token gate	1,415,180	46.0	39
Multi-scale fusion + token gate, CG blocks	1,306,724	151.1	8
Multi-scale fusion + token gate, shared weights	392,198	45.8	32
Mask-prompt generator (segmentation loss)	671,138	3.7	29
Mask-prompt generator (end-to-end via SAM)	671,138	26.4	23
Mask-prompt generator (two-stage)	671,138	12.4	4
Mask-prompt generator (segmentation loss, label smoothing)	671,138	4.2	29
Mask-prompt generator + raw CG-Net fields	686,306	5.9	29
Box head on SAM features (YOLO-style)	608,778	68.6	20
Learned static prompts (4 tokens / class)	2,048	12.6	48
Learned static prompts (16 tokens / class)	8,192	14.2	48

Table A.4: Size and training cost (single A100, cached encoder features, 60 epochs; two-stage: 30 extra epochs on top of the segmentation-loss model).

Prompter	Output	TC IoU	AR IoU	BG IoU	Mean IoU	Mean FG IoU
Ground truth (oracle)	Prompter mask (no SAM)	1.000	1.000	1.000	1.000	1.000
	SAM + box	0.728	0.650	0.973	0.784	0.689
	SAM + hybrid	0.857	0.905	0.993	0.918	0.881
CG-Net (official weights)	Prompter mask (no SAM)	0.327	0.333	0.926	0.529	0.330
	SAM + box	0.318	0.342	0.931	0.530	0.330
	SAM + hybrid	0.328	0.339	0.929	0.532	0.333
CG-Net (fine-tuned)	Prompter mask (no SAM)	0.349	0.382	0.940	0.557	0.366
	SAM + box	0.341	0.375	0.940	0.552	0.358
	SAM + hybrid	0.348	0.384	0.941	0.558	0.366
Logistic regression, ViT block 1	Prompter mask (no SAM)	0.185±0.001	0.305±0.001	0.924±0.003	0.471±0.002	0.245±0.001
	SAM + box	0.186±0.002	0.312±0.003	0.925±0.002	0.474±0.000	0.249±0.001
	SAM + hybrid	0.186±0.001	0.310±0.001	0.928±0.003	0.474±0.001	0.248±0.001
Logistic regression, ViT block 12	Prompter mask (no SAM)	0.295±0.004	0.388±0.002	0.941±0.001	0.542±0.002	0.342±0.003
	SAM + box	0.297±0.002	0.377±0.001	0.937±0.001	0.537±0.002	0.337±0.002
	SAM + hybrid	0.300±0.004	0.367±0.006	0.946±0.001	0.538±0.003	0.333±0.004
Multi-scale fusion	Prompter mask (no SAM)	0.330	0.409	0.943	0.561	0.370
	SAM + box	0.337	0.397	0.941	0.558	0.367
	SAM + hybrid	0.337	0.404	0.946	0.562	0.370
Multi-scale fusion + token gate	Prompter mask (no SAM)	0.319	0.404	0.941	0.555	0.362
	SAM + box	0.327	0.390	0.938	0.552	0.358
	SAM + hybrid	0.329	0.404	0.945	0.559	0.367
Mask-prompt generator (segmentation loss)	Prompter mask (no SAM)	0.330±0.010	0.410±0.005	0.944±0.003	0.561±0.004	0.370±0.008
	SAM + box	0.333±0.008	0.395±0.003	0.941±0.003	0.557±0.003	0.364±0.006
	SAM + hybrid	0.332±0.011	0.404±0.010	0.947±0.002	0.561±0.006	0.368±0.010
Mask-prompt generator (end-to-end via SAM)	Prompter mask (no SAM)	0.323±0.009	0.350±0.015	0.947±0.001	0.540±0.007	0.336±0.012
	SAM + box	0.328±0.008	0.383±0.005	0.943±0.003	0.552±0.003	0.356±0.006
	SAM + hybrid	0.328±0.009	0.357±0.024	0.951±0.001	0.546±0.010	0.343±0.016
Mask-prompt generator (two-stage)	Prompter mask (no SAM)	0.326±0.005	0.410±0.001	0.940±0.002	0.559±0.002	0.368±0.003
	SAM + box	0.333±0.007	0.393±0.004	0.937±0.002	0.554±0.004	0.363±0.005
	SAM + hybrid	0.331±0.007	0.413±0.002	0.945±0.001	0.563±0.002	0.372±0.003
Mask-prompt generator (segmentation loss, label smoothing)	Prompter mask (no SAM)	0.334±0.004	0.409±0.003	0.945±0.001	0.563±0.002	0.371±0.002
	SAM + box	0.338±0.008	0.398±0.002	0.943±0.001	0.560±0.003	0.368±0.004
	SAM + hybrid	0.336±0.004	0.403±0.004	0.948±0.000	0.562±0.001	0.369±0.002

Table A.5: Automatic prompting on the ClimateNet test set (61 images), frozen ClimateSAM (Infused Token, MLP ratio 0.5). Learned prompts: mean±std over 3 seeds, checkpoint selected on 40 held-out training images.

Prompter, prompts, training prompts	Output	TC IoU	AR IoU	Mean FG IoU	TC recall / prec.	AR recall / prec.
Mask-prompt generator, bbox, out-of-fold	prompter mask	0.337	0.414	0.376		
	SAM, Phase-1 decoder	0.340	0.399	0.369	0.62 / 0.55	0.95 / 0.56
	SAM, adapted decoder	0.316	0.390	0.353	0.63 / 0.52	0.91 / 0.62
Mask-prompt generator, bbox, in-sample	prompter mask	0.337	0.414	0.376		
	SAM, Phase-1 decoder	0.340	0.399	0.369	0.62 / 0.55	0.95 / 0.56
	SAM, adapted decoder	0.340	0.404	0.372	0.63 / 0.56	0.93 / 0.64
Mask-prompt generator, hybrid, out-of-fold	prompter mask	0.337	0.414	0.376		
	SAM, Phase-1 decoder	0.339	0.413	0.376	0.62 / 0.56	0.92 / 0.63
	SAM, adapted decoder	0.338	0.416	0.377	0.63 / 0.55	0.94 / 0.62

Table A.6: Adapting the HQ decoder parts to generated prompts (test set). Phase-1 decoder = the frozen decoder of all other experiments; adapted = best validation epoch of 30.

Prompt type	Pos.	Neg.	Enlarge	IoU AR	IoU TC	IoU AR (orig. script)	IoU TC (orig. script)
CG-Net alone (baseline)				0.3817	0.3494		
point+bbox	10	5	0.0	0.3602	0.3447	0.3560	0.3430
bbox	0	0	0.0	0.3750	0.3423	0.3758	0.3404
point+bbox	15	10	0.0	0.3524	0.3420	0.3490	0.3399
bbox	0	0	-0.1	0.3327	0.3322	0.3355	0.3310
bbox	0	0	0.1	0.3625	0.3122	0.3572	0.3109
point	10	10	0.0	0.3376	0.3054	0.3297	0.3060
point	1	3	0.0	0.2856	0.3016	0.2859	0.3001
point	5	5	0.0	0.3425	0.2996	0.3380	0.3047
point	5	10	0.0	0.3402	0.2980	0.3342	0.3065
point	1	1	0.0	0.3224	0.2970	0.3214	0.2954
point	2	2	0.0	0.3260	0.2910	0.3226	0.2854
point	1	2	0.0	0.3013	0.2887	0.3009	0.2840
bbox	0	0	0.2	0.3343	0.2661	0.3260	0.2645
bbox	0	0	-0.2	0.2211	0.2494	0.2271	0.2523
bbox	0	0	0.3	0.3026	0.2279	0.2918	0.2268
bbox	0	0	0.5	0.2437	0.1597	0.2297	0.1613
point+bbox+mask	20	10	0.0	0.2352	0.1296	0.2469	0.1292
point+bbox+mask	10	5	0.0	0.2316	0.1129	0.2349	0.1022
mask	0	0	0.0	0.3794	0.0000	0.3724	0.0000

Table A.7: SAM with CG-Net as prompter, re-run on the 61 test images (Infused Token MLP 1.0, trained without label smoothing). “orig. script”: image passed through `ClimateSAM.set_infer_img`, which skips the learned input adapter (`test_prompt_effect.py`); other columns: input adapter applied as in training. Ordered by TC IoU.

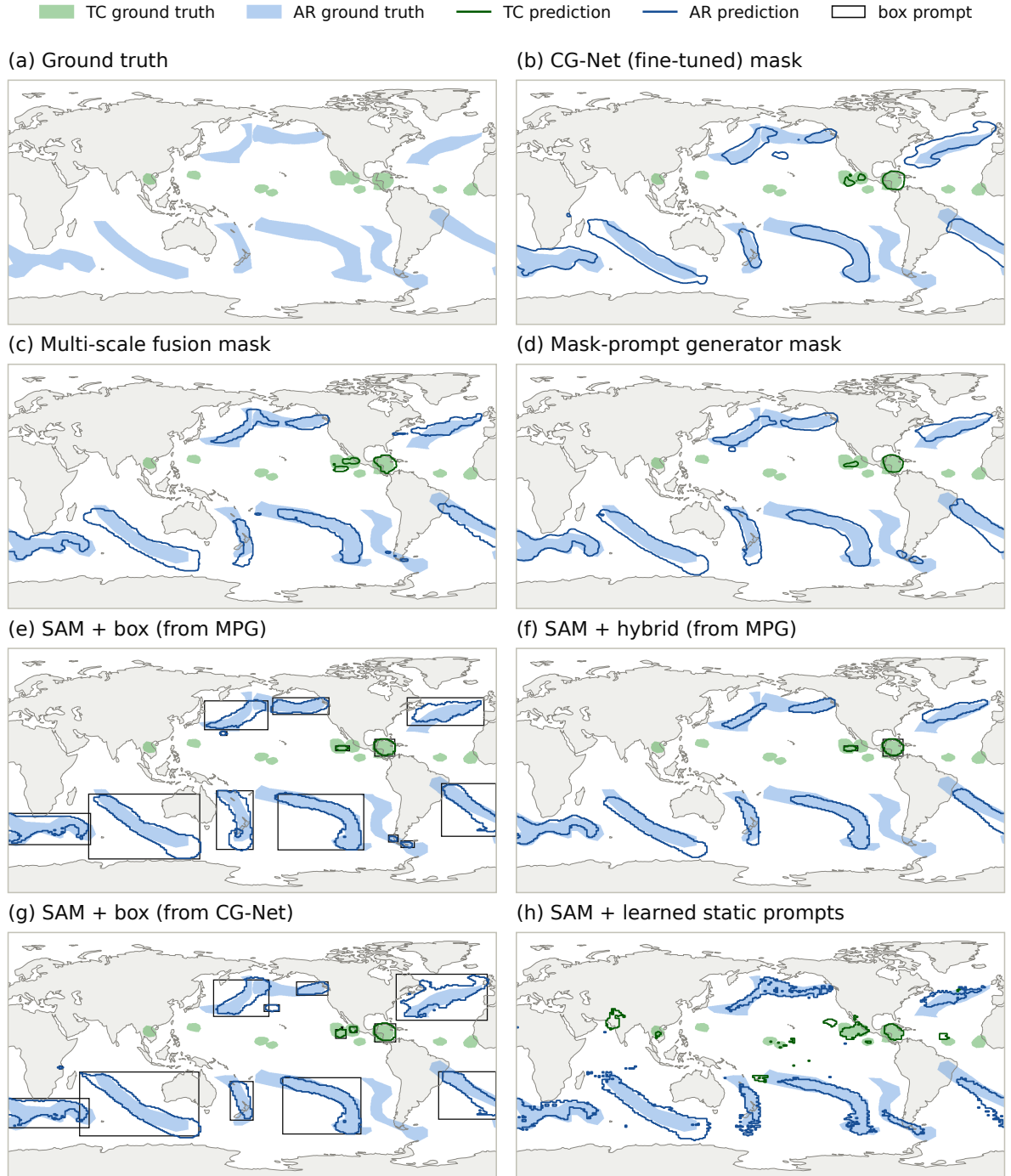


Figure A.1: Test image 45 (layout as in Figure 4.15).

7. The fine-tuned CG-Net was trained on all 398 training images, so its outputs on training images are optimistic.
8. The encoder file used by most Phase-2 prompter runs was overwritten after these runs, and two checkpoint files named after 256-channel runs contain 128-channel models (Section A.3).

A.3 Re-evaluation of the Chapter 4 Results

The saved checkpoints of Chapter 4 were evaluated again with the thesis’ own evaluation pipeline: the ClimateNet test images with prompts derived from the ground truth (box, point or a random choice per image), `ClimateSAM.infer`, and TC and AR scored as two independent binary problems. Available were the four Infused Token models of the appendix table (linear and nonlinear input adapter, MLP ratio 1.0 and 0.5) and all Phase-2 prompter checkpoints; the Concat Token, Single Token and LoRA checkpoints are not available, and for Single Token and Single LoRA the training logs were used instead.

A.3.1 The Phase-1 validation skips the input adapter

The validation routine of the Phase-1 training script encodes the images with `ClimateSAM.set_infer_img()`, which passes the first three raw channels (TMQ, U850, V850) to the encoder and does not apply the learned input adapter; training uses `encode_images()`, which does. Table A.8 and Figure A.2 compare the reported values with a re-run of the original evaluation and with a re-run in which the adapter is applied.

Infused Token	Prompt	Thesis TC	Thesis AR	As in thesis TC	As in thesis AR	Adapter applied TC	Adapter applied AR	Δ FG (applied – thesis)
linear, MLP 1.0	box	0.724	0.632	0.730	0.616	0.736	0.625	+0.002
	point	0.522	0.456	0.525	0.442	0.532	0.453	+0.003
	random	0.595	0.523	0.624 (0.613–0.642)	0.521 (0.521–0.521)	0.637 (0.622–0.657)	0.530 (0.529–0.532)	+0.025
linear, MLP 0.5	box	0.724	0.645	0.738	0.648	0.741	0.650	+0.011
	point	0.526	0.469	0.526	0.470	0.533	0.486	+0.012
	random	0.606	0.536	0.627 (0.612–0.645)	0.555 (0.554–0.556)	0.636 (0.621–0.653)	0.560 (0.558–0.561)	+0.027
nonlinear, MLP 1.0	box	0.699	0.445	0.733	0.439	0.721	0.556	+0.067
	point	0.331	0.313	0.394	0.314	0.520	0.440	+0.158
	random	0.447	0.360	0.523 (0.513–0.541)	0.356 (0.349–0.364)	0.623 (0.606–0.643)	0.491 (0.486–0.496)	+0.153
nonlinear, MLP 0.5	box	0.733	0.522	0.749	0.536	0.748	0.622	+0.057
	point	0.476	0.351	0.461	0.373	0.531	0.462	+0.083
	random	0.572	0.421	0.613 (0.591–0.638)	0.442 (0.439–0.447)	0.642 (0.617–0.673)	0.530 (0.528–0.534)	+0.090

Table A.8: Re-evaluation of the Phase-1 checkpoints (appendix Table 1) on the 61 test images with the thesis pipeline. ”As in thesis”: images encoded with `set_infer_img` (first three raw channels, learned input adapter skipped), as in the validation of `train_adaptation.py`; ”adapter applied”: `encode_images`, as in training. Random prompts: mean of three prompt draws (range).

For the linear adapter the reported box and point values reproduce within about ± 0.015 IoU, and applying the adapter changes them by at most 0.02: the adapter keeps its identity weights for TMQ, U850 and V850 (Table 4.7), so the raw channels have the same spatial structure as its output. The nonlinear adapter, in contrast, maps the inputs to low-contrast images (Figure A.3); its encoder was trained on these images but evaluated on the raw channels. With the adapter applied, the nonlinear models gain 0.04–0.13 mean FG IoU and 0.09–0.14 AR IoU, and the advantage of the linear adapter shrinks from 0.06–0.17 to 0.005–0.042 mean FG IoU (Table A.9), with the remaining gain almost entirely in AR. The large gap of Table 4.6 is therefore caused by the evaluation and not by spatial blurring in the nonlinear adapter. Figure A.4 shows the effect on one test image.

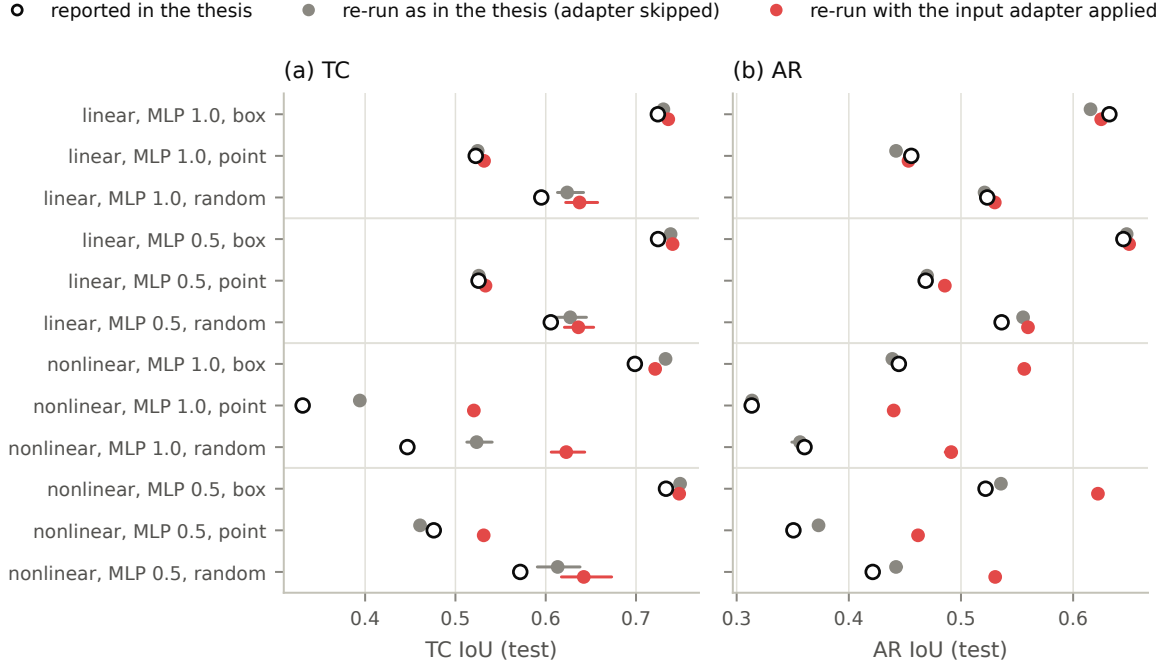


Figure A.2: Phase-1 checkpoints with ground-truth prompts: value reported in Table 4.12 / appendix Table 1 (ring), re-run with the original evaluation (grey, input adapter skipped) and with the input adapter applied (red). Random prompts: mean and range over three prompt draws.

Infused Token	Prompt	Comparison (B – A)	Δ TC [95% CI]	Δ AR [95% CI]	Δ mean FG [95% CI]
MLP 1.0	box	nonlinear: adapter applied vs. skipped	-0.012 [-0.021, -0.002]	+0.118 [+0.097, +0.137]	+0.053 [+0.042, +0.064]
MLP 1.0	box	linear vs. nonlinear, as in the thesis	-0.002 [-0.010, +0.005]	+0.177 [+0.158, +0.193]	+0.087 [+0.077, +0.096]
MLP 1.0	box	linear vs. nonlinear, adapter applied	+0.015 [+0.006, +0.024]	+0.069 [+0.056, +0.081]	+0.042 [+0.035, +0.048]
MLP 1.0	point	nonlinear: adapter applied vs. skipped	+0.126 [+0.102, +0.150]	+0.126 [+0.114, +0.138]	+0.126 [+0.112, +0.140]
MLP 1.0	point	linear vs. nonlinear, as in the thesis	+0.131 [+0.111, +0.152]	+0.128 [+0.114, +0.142]	+0.129 [+0.118, +0.142]
MLP 1.0	point	linear vs. nonlinear, adapter applied	+0.011 [-0.004, +0.027]	+0.013 [+0.003, +0.025]	+0.012 [+0.003, +0.023]
MLP 1.0	random	nonlinear: adapter applied vs. skipped	+0.103 [+0.060, +0.144]	+0.143 [+0.128, +0.157]	+0.123 [+0.099, +0.145]
MLP 1.0	random	linear vs. nonlinear, as in the thesis	+0.101 [+0.063, +0.139]	+0.172 [+0.153, +0.190]	+0.137 [+0.115, +0.156]
MLP 1.0	random	linear vs. nonlinear, adapter applied	+0.014 [+0.000, +0.026]	+0.037 [+0.025, +0.050]	+0.026 [+0.016, +0.034]
MLP 0.5	box	nonlinear: adapter applied vs. skipped	-0.001 [-0.007, +0.005]	+0.087 [+0.076, +0.098]	+0.043 [+0.036, +0.049]
MLP 0.5	box	linear vs. nonlinear, as in the thesis	-0.010 [-0.018, -0.003]	+0.112 [+0.098, +0.128]	+0.051 [+0.043, +0.059]
MLP 0.5	box	linear vs. nonlinear, adapter applied	-0.007 [-0.014, -0.000]	+0.028 [+0.018, +0.037]	+0.010 [+0.004, +0.016]
MLP 0.5	point	nonlinear: adapter applied vs. skipped	+0.070 [+0.038, +0.108]	+0.089 [+0.077, +0.101]	+0.080 [+0.062, +0.099]
MLP 0.5	point	linear vs. nonlinear, as in the thesis	+0.065 [+0.028, +0.107]	+0.097 [+0.083, +0.110]	+0.081 [+0.060, +0.103]
MLP 0.5	point	linear vs. nonlinear, adapter applied	+0.002 [-0.016, +0.018]	+0.024 [+0.015, +0.034]	+0.013 [+0.002, +0.023]
MLP 0.5	random	nonlinear: adapter applied vs. skipped	+0.035 [+0.013, +0.058]	+0.090 [+0.077, +0.102]	+0.062 [+0.049, +0.075]
MLP 0.5	random	linear vs. nonlinear, as in the thesis	+0.007 [-0.024, +0.039]	+0.118 [+0.098, +0.137]	+0.062 [+0.044, +0.081]
MLP 0.5	random	linear vs. nonlinear, adapter applied	-0.020 [-0.038, -0.005]	+0.030 [+0.018, +0.043]	+0.005 [-0.005, +0.014]

Table A.9: Linear versus nonlinear input adapter (Table 4.6) and the effect of the skipped adapter, paired bootstrap over the 61 test images (random prompts: first draw, identical for both models).

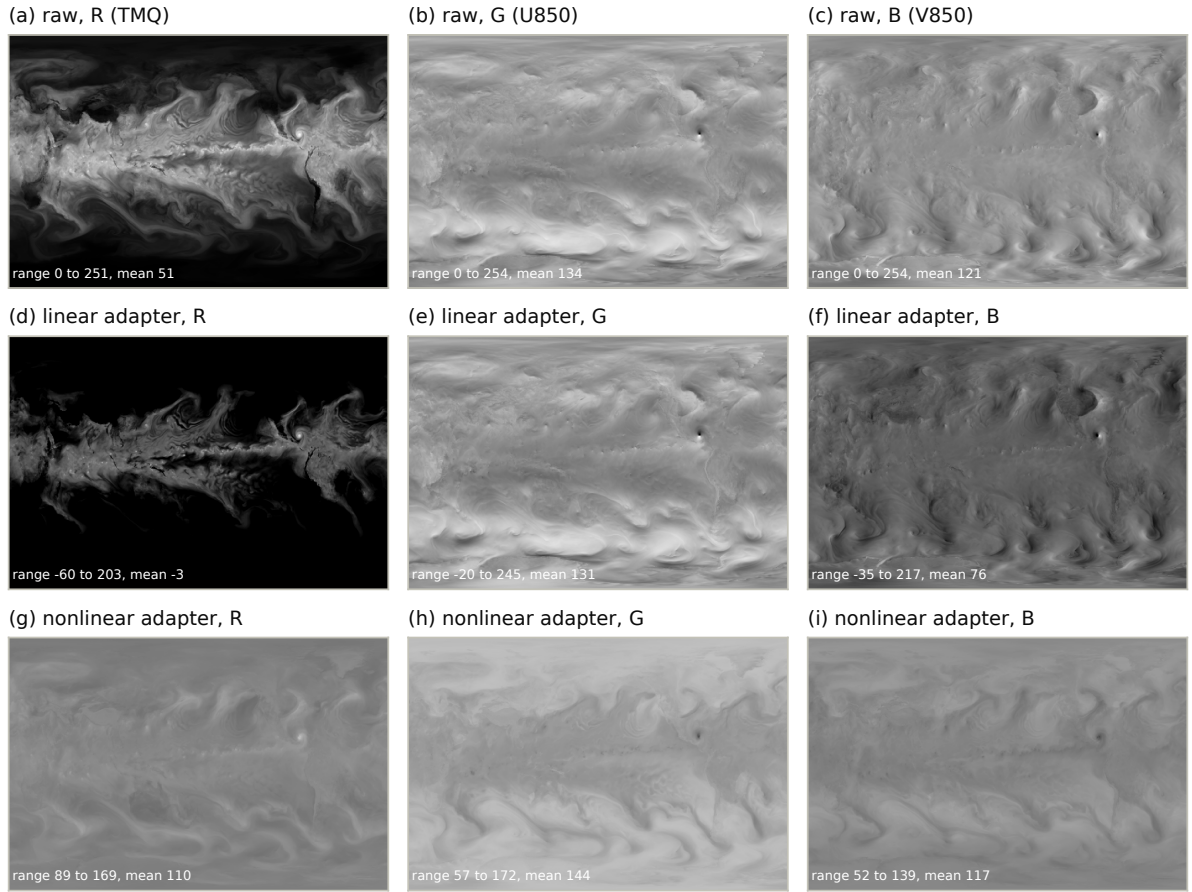
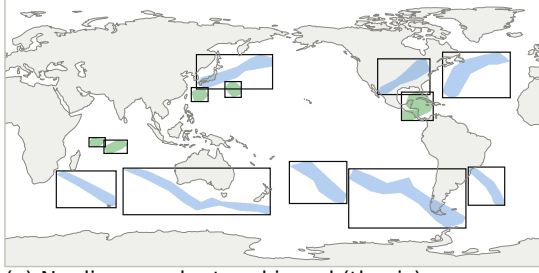


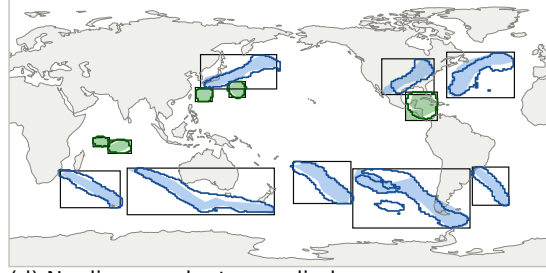
Figure A.3: Input of the image encoder for test image 17: the raw channels used at evaluation (a–c) and the outputs of the linear (d–f) and nonlinear (g–i) input adapters on which the respective encoders were trained (grey scale 0–255).

TC ground truth AR ground truth TC prediction AR prediction box prompt (ground truth)

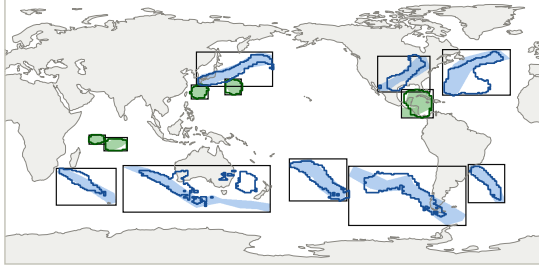
(a) Ground truth and box prompts



(b) Linear, as in the thesis



(c) Nonlinear, adapter skipped (thesis)



(d) Nonlinear, adapter applied

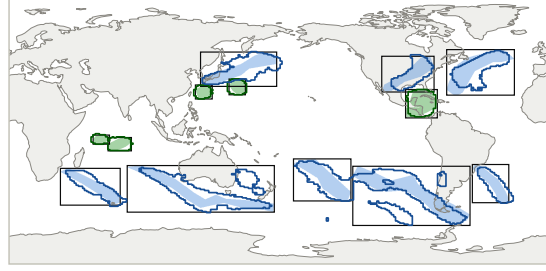


Figure A.4: Infused Token (MLP ratio 1.0) with ground-truth box prompts on test image 17: linear adapter as evaluated in the thesis (b), nonlinear adapter as evaluated in the thesis (c) and with its adapter applied (d).

A.3.2 Checkpoint selection on the test set

The Phase-1 training script validates on the 61 test images every five epochs and saves the epoch with the best mean of TC and AR IoU. Figure A.5 shows the logged test IoU of the training runs. For the token-based adapters the curves plateau and the selected epoch differs from the last one by at most 0.01 IoU. For Single LoRA the TC IoU collapses during training, from 0.63 at epoch 6 to 0.10 at epoch 50 for rank 32 and from 0.59 to 0.12 for rank 64; the LoRA values of Tables 4.6 and 4.8 therefore describe the best early test epoch, and both ranks, not only rank 64, are unstable for TCs.

Run (wandb)	Max TC IoU	Max AR IoU	Best mean epoch TC / AR	Last epoch TC / AR
infused_token_vit_b_mlp1_CORRECTED	0.648 (ep. 46)	0.558 (ep. 46)	0.648 / 0.558 (ep. 46)	0.641 / 0.556 (ep. 50)
infused_token_vit_b_mlp05_CORRECTED	0.647 (ep. 46)	0.595 (ep. 46)	0.647 / 0.595 (ep. 46)	0.644 / 0.594 (ep. 50)
single_sam_vit_b_CORRECTED	0.644 (ep. 46)	0.548 (ep. 50)	0.644 / 0.546 (ep. 46)	0.641 / 0.548 (ep. 50)
lora_single_vit_b_r32_CORRECTED	0.634 (ep. 6)	0.594 (ep. 21)	0.634 / 0.568 (ep. 6)	0.104 / 0.579 (ep. 50)
lora_single_vit_b_r64_CORRECTED	0.592 (ep. 11)	0.591 (ep. 16)	0.592 / 0.569 (ep. 11)	0.116 / 0.588 (ep. 50)
infused_token_vit_b_mlp1_nonlinear	0.478 (ep. 46)	0.352 (ep. 41)	0.477 / 0.352 (ep. 41)	0.477 / 0.344 (ep. 50)
infused_token_vit_b_mlp05_nonlinear	0.586 (ep. 50)	0.424 (ep. 46)	0.586 / 0.424 (ep. 46)	0.586 / 0.422 (ep. 50)
single_sam_vit_b_nonlinear	0.589 (ep. 46)	0.451 (ep. 36)	0.587 / 0.449 (ep. 41)	0.589 / 0.447 (ep. 50)
lora_single_vit_b_r32_nonlinear	0.481 (ep. 11)	0.506 (ep. 16)	0.481 / 0.458 (ep. 11)	0.034 / 0.496 (ep. 50)
lora_single_vit_b_r64_nonlinear	0.530 (ep. 6)	0.514 (ep. 21)	0.530 / 0.487 (ep. 6)	0.037 / 0.505 (ep. 50)
infused_token_vit_b_mlp1_CORRECTED_NOSMOOTH	0.661 (ep. 31)	0.583 (ep. 50)	0.657 / 0.583 (ep. 50)	0.657 / 0.583 (ep. 50)

Table A.10: Test-set IoU logged during Phase-1 training (the author’s wandb runs; validation = the 61 test images, every 5 epochs, input adapter skipped). The saved checkpoint is the epoch with the best mean of TC and AR.

A.3.3 Further checks

The random-prompt rows cannot be reproduced exactly, because every evaluation draws a new mixture of point and box prompts (TC spread of 0.03 across three draws). The adapter weights of Table 4.7 are those of the run without label smoothing, not of the smoothed run whose

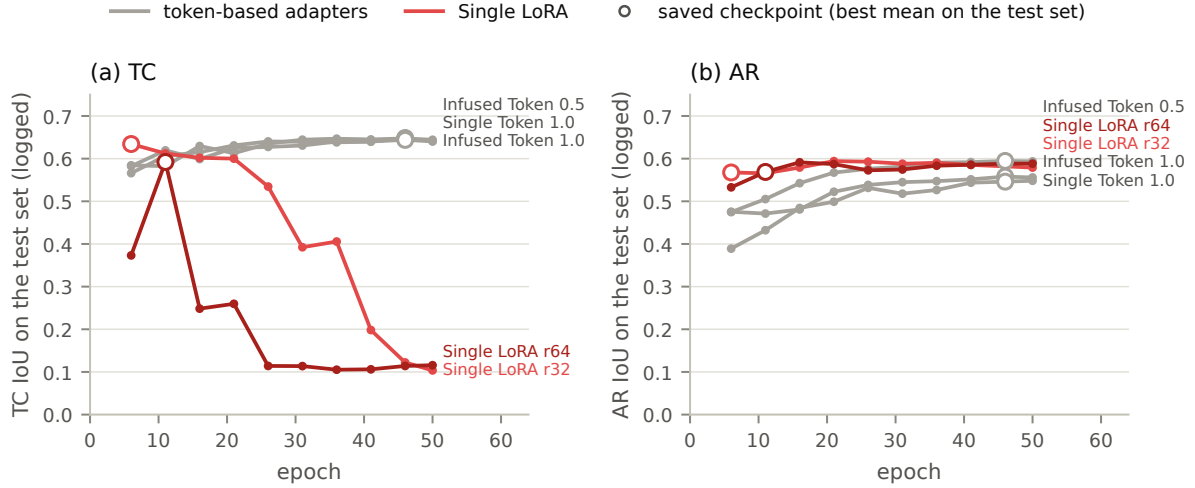


Figure A.5: Test IoU logged during the Phase-1 training runs (every five epochs); ring: saved checkpoint.

IoUs Figure 4.2 quotes. The parameter counts of Table 4.11 are exact for all Infused Token configurations.

A.3.4 Phase-2 prompter checkpoints

The prompter checkpoints of Section 4.3.3 were evaluated with the encoder each was trained with and with their own decision rule (three-class argmax), Table A.11 and Figure A.6. The multi-scale fusion run whose encoder is unchanged reproduces its logged values exactly (mask TC 0.304 / AR 0.401 versus 0.305 / 0.402). The other generators were trained on an encoder file that was overwritten after the runs; with the current file their TC IoU is far below the logged values, so they cannot be verified. Where they can be evaluated, SAM prompted by them is not better than their masks, and the same architectures trained under the protocol of Section 3.6 are equal or better.

Prompter checkpoint (run)	Prompts	Logged mask (best)	Re-run mask	Logged SAM (best)	Re-run SAM	Same arch. retrained
MSF (Generator)	point	- / -	0.251 / 0.310	- / -	0.158 / 0.223	0.333 / 0.403
MSF (generator_128_vit_b_bbox)	box	0.305 / 0.402	0.304 / 0.401	0.308 / 0.389	0.294 / 0.383	0.333 / 0.403
MSF + token gate (FIRST)	point	0.321 / 0.056	0.256 / 0.056	0.304 / 0.291	0.295 / 0.074	0.324 / 0.399
token gate + CG blocks (CG)	point	0.349 / 0.407	0.157 / 0.374	0.329 / 0.346	0.274 / 0.291	0.324 / 0.355
token gate + CG blocks (CG_256)	box	0.315 / 0.375	0.144 / 0.360	0.318 / 0.331	0.175 / 0.373	0.324 / 0.355
token gate + CG blocks (CG_256_REAL)	box	0.327 / 0.387	0.118 / 0.352	0.322 / 0.332	0.141 / 0.371	0.324 / 0.355
token gate, shared weights (SP_TOKEN)	box	0.326 / 0.393	0.205 / 0.376	0.333 / 0.306	0.190 / 0.361	0.332 / 0.402
logistic regression, block 1 (Logistic)	box	0.191 / 0.295	0.190 / 0.287	0.183 / 0.255	0.194 / 0.298	0.195 / 0.319

Table A.11: The author’s Phase-2 prompter checkpoints (TC IoU / AR IoU, 61 test images), each with the frozen encoder it was trained with. Logged: best value in the author’s wandb run (selected on the test set; the run logged SAM with its own prompt pipeline). Re-run mask: 3-class argmax, as in the training scripts. Same arch. retrained: the architecture trained under the benchmark protocol (two sigmoid channels, selection on validation, primary checkpoint, mean of seeds). SAM hybrid and object counts: 08_thesis_recheck/phase2_all.csv.

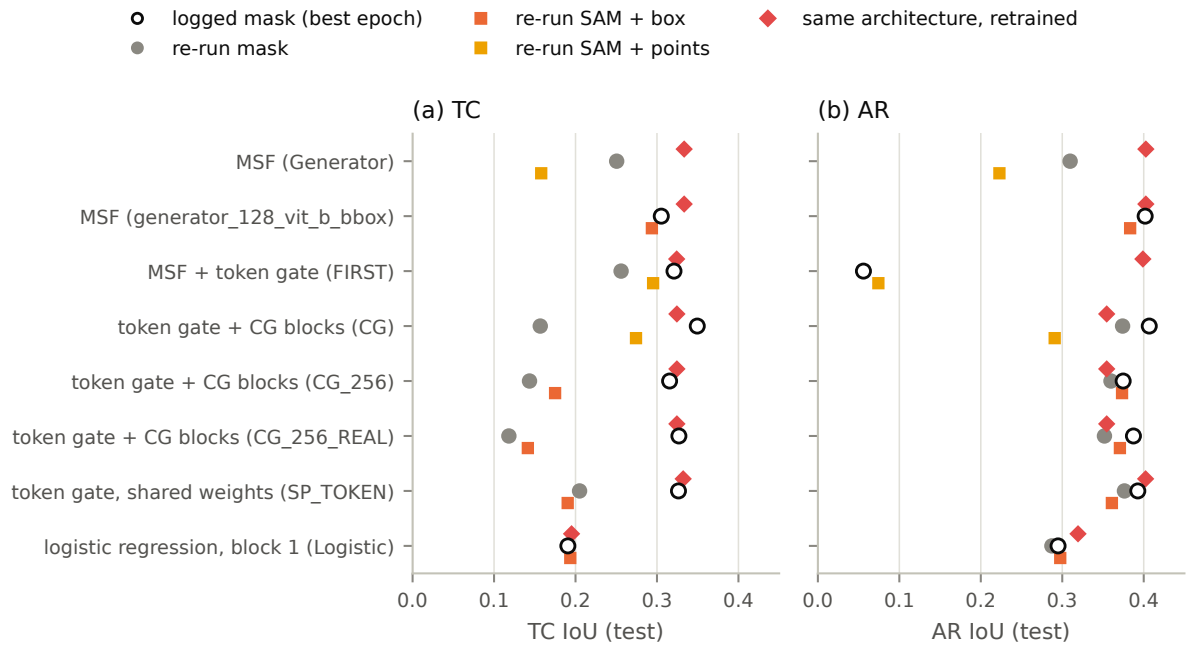


Figure A.6: The Phase-2 prompt checkpoints of the thesis: best logged value (ring), re-run mask (grey), re-run SAM with the prompt type used in training (squares), and the same architecture trained under the benchmark protocol (red).

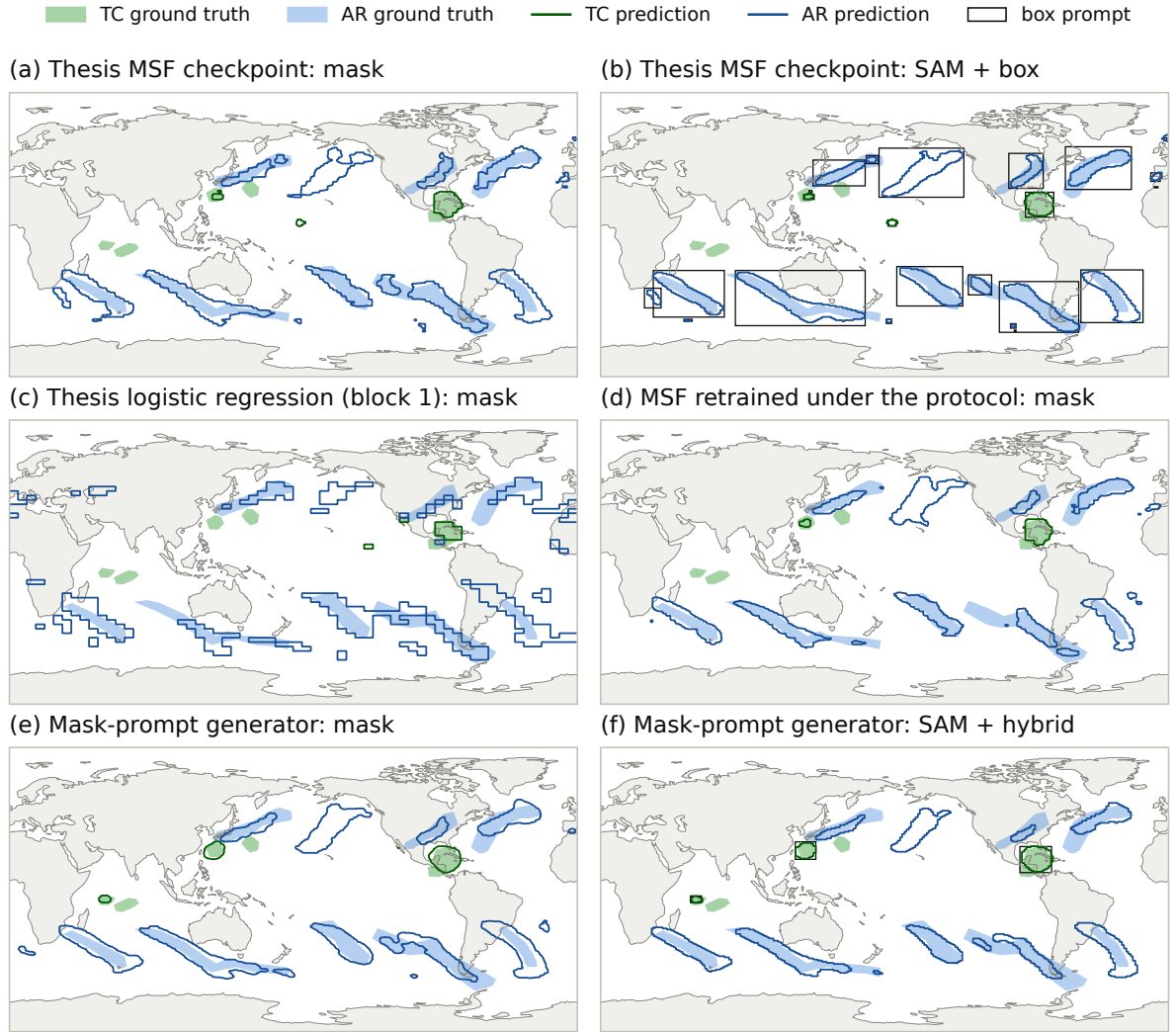


Figure A.7: Test image 17: the thesis’ multi-scale fusion checkpoint (a) and SAM prompted by its boxes (b), the thesis’ logistic regression on ViT block 1 (c), and multi-scale fusion (d) and the mask-prompt generator (e, f) trained under the benchmark protocol.